Cross-Modal Contrastive Learning With Spatiotemporal Context for Correlation-Aware Multiscale Remote Sensing Image Retrieval

Lilu Zhu, Yang Wang, *Member, IEEE*, Yanfeng Hu, Xiaolu Su, and Kun Fu, *Member, IEEE*

***Abstract*— Optical satellites are the most popular observation platforms for humans viewing Earth. Driven by rapidly developing multisource optical remote sensing technology, content-based remote sensing image retrieval (CBRSIR), which aims to retrieve images of interest using extracted visual features, faces new challenges derived from large data volumes, complex feature information, and various spatiotemporal resolutions. Most previous works delve into optical image representation and transformation to the semantic space of retrieval via supervised or unsupervised learning. These retrieval methods fail to fully leverage geospatial information, especially spatiotemporal features, which can improve the accuracy and efficiency to some extent. In this article, we propose a cross-modal contrastive learning method (CCLS2T) to maximize the mutual information of multisource remote sensing platforms for correlation-aware retrieval. Specifically, we develop an asymmetric dual-encoder architecture with a vision encoder that operates on multiscale visual inputs, and a lightweight text encoder that reconstructs spatiotemporal embeddings and adopts an intermediate contrastive objective on representations from unimodal encoders. Then, we add a hash layer to transform the deep fusion features into compact hash index codes. In addition, CCLS2T exploits the prompt template (R2STFT) for multisource remote sensing retrieval to address the text heterogeneity of metadata files and the hierarchical semantic tree (RSHST) to address the feature sparsification of semantic-aware indexing structures. The experimental results on three optical remote sensing datasets substantiate that the proposed CCLS2T can improve retrieval performance by 11.64% and 9.91% compared with many existing hash learning methods and server-side retrieval engines, respectively, in typical optical remote sensing retrieval scenarios.**

***Index Terms*— Content-based remote sensing image retrieval (CBRSIR), correlation-aware retrieval, cross-modal contrastive learning, hash index code, hierarchical semantic tree.**

I. Introduction

With the vigorous expansion of Earth observation and sensor techniques, there is obvious multisource heterogeneity in current optical remote sensing images in terms of acquisition platforms, scales, and features. While optical remote sensing images are susceptible to the influences of clouds, the atmosphere, light, and other natural climatic factors, they remain advantageous in a multitude of ways. High-resolution optical remote sensing images offer clear spatial texture information about the observed targets and characterize the nature of features, and they are employed in a wide range of fields. Most conventional similarity retrieval techniques utilizing low-level characteristics are severely constrained in terms of their ability to satisfy broad suitability and applicability criteria, which are aimed at the identification and retrieval of remote sensing images of interest from an extensive image corpus. At present, we analyze the challenge of multisource remote sensing image retrieval (MSRSIR) in three phases. First, for both natural images and optical remote sensing images, most users intuitively judge and distinguish the final retrieval results by relying on specific image semantics rather than low-level visual features. For example, we want to retrieve rivers and lakes from images instead of qualified histograms and matrices. The difference between the image feature space generated by a model and the high-level semantic space understood by people is called the "semantic gap" [1], which decreases the precision of image retrieval (IR), as shown in Fig. 1(a). Second, the changes in surface features and the complexity of the geographical environment recorded by remote sensing images also lead to the absence of unique image elements. Experts may derive different conclusions for the same remote sensing scene from their research perspectives. Therefore, neither content-based IR methods nor semantic-based retrieval methods for natural images apply to remote sensing images. Third, it is common sense that optical remote sensing images of the same geographic region but at different scales will have different visual features; that is, with changes in scale or resolution, the textural and shape details in the same patch exhibit obvious differences, as shown in Fig. 1(b). Furthermore, there are also common problems, such as "same object with different spectra" and "different objects with the same spectrum," in multisource optical remote sensing images; these problems need to be investigated thoroughly in the future.

In general, IR refers to the extraction of images or image sequences that are relevant to a query from a storage system such as an image database. Most early IR methods involve text retrieval, which converts IR into a keyword retrieval process through manual annotation. However, text-based IR (TBIR) [2], [3] methods rely on many text tags and are limited by the background and direction of the individual or

Manuscript received 17 January 2024; revised 17 May 2024; accepted 15 June 2024. Date of publication 21 June 2024; date of current version 5 July 2024. This work was supported by Suzhou Cutting-Edge Technology Research Project under Grant SYG202335. *(Corresponding author: Lilu Zhu.)*

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Digital Object Identifier 10.1109/TGRS.2024.3417421

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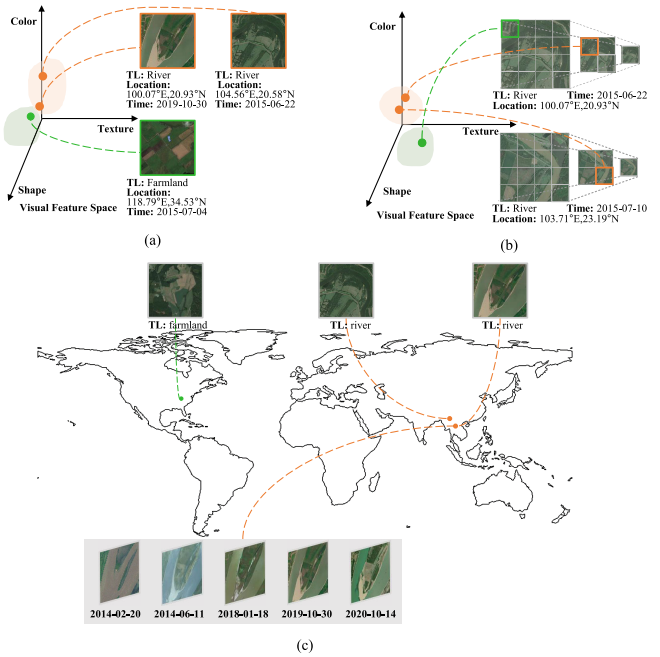


Fig. 1. Intractable problems exist in optical remote sensing IR and corresponding solutions. (a) Feature homogeneity of multisource optical remote sensing images. (b) Feature heterogeneity of multiscale optical remote sensing images. (c) Optical remote sensing images with different STCs.

group being researched. Some researchers transfer TBIR to the field of remote sensing, resulting in text-based remote sensing IR (TBRSIR) [4], [5]. In addition to the issues encountered in TBIR, the diverse feature elements and complex spatial relationships contained in remote sensing images cannot be objectively described in full by pure text annotations, which weaken the final retrieval results. In contrast, content-based remote sensing IR (CBRSIR) utilizes visual features extracted from images in similarity calculations to achieve IR without much manual tagging. Nevertheless, the calculation of feature similarities among multisource optical remote sensing images with high resolutions or ultrahigh-resolutions requires a large amount of high-performance computing resources, which greatly constrains the efficiency of CBRSIR [6], [7], [8], [9], [10]. However, as mentioned earlier, there is a large discrepancy between visual features and our understanding, namely, the high-level semantics of images. However, accurately expressing the evolution of the data scale and spatiotemporal relationships with a global feature is difficult, and additional implicit semantic features need to be found to improve the accuracy of CBRSIR. In contrast to natural and medical images, optical remote sensing images have abundant geographic attributes, including spatial, temporal, platform, and domain attributes. To address the problems mentioned in Fig. 1(a) and (b), we find that some prior spatiotemporal contexts (STCs) can help significantly alleviate this issue. As shown in Fig. 1(c), two optical remote sensing images with similar visual representations can be easily distinguished by their spatial distributions and temporal attributes. In summary, through mining and merging the prior geographic information in multisource optical remote sensing images, the expression of the feature space can be further optimized, thus improving the performance of downstream tasks [11], [12], [13], [14], [15], [16], [17].

For many heterogeneous optical remote sensing images, attaining a semantic-based retrieval model via the formalization of feature relationships and geospatial cognition is an extremely difficult task. Therefore, different from object recognition and semantic segmentation, we aim to achieve semantic-based hash retrieval using potential semantic correlations in heterogeneous data according to the characteristics of retrieval tasks. Alternatively, the discrimination of remote sensing semantic correlations should receive more attention instead of remote sensing semantic descriptions. For this purpose, we refer to Robert Epstein's experiment [18], where he encouraged experimenters to draw representations of a U.S. $1 bill by either drawing from memory or copying it on the spot. Although there are significant differences between the drawings generated through the two methods, people can comprehend that they depict the same target, the U.S. $1 bill, because they have communal representations, such as Washington's figure, the one-dollar caption, and others. Inspired by this conclusion, we can construct a representation learning model that not only focuses on pixel-level details but is also sufficient for capturing potential semantically related information from optical remote sensing images to improve CBRSIR. Multimodal learning [19] can learn better feature representations from multisource data by exploiting complementary information. In contrast, contrastive learning [20] enables the model to learn more hidden distinguishing features, which helps similar samples approach each other more closely and dissimilar samples move away. Accordingly, multimodal contrastive learning [21] is not a new paradigm, and pretraining with the contrastive loss is a major breakthrough in the computer vision domain because of its formidable feature extraction and expression ability [22], [23], [24], [25], [26], [27]. Consequently, based on the rich metadata text of remote sensing data, we construct a semantic correlation-aware retrieval model through feature representation and contrastive analysis of two-modal datasets.

In this article, we redefine CBRSIR into a discriminative learning task of correlation semantic indexing by incorporating existing geographical prior knowledge into multimodal contrastive learning. To this end, we propose a new cross-modal contrastive learning method (CCLS2T) for correlation analysis and index generation in multisource optical remote sensing. In contrast to previous proposals, CCLS2T delves into the correlations of both visual and spatiotemporal semantics simultaneously rather than as a global visual feature and maps them into a unified hash-based index space. In more detail, we develop an asymmetric dual-encoder architecture for optical remote sensing images and metadata files, with a vision encoder constructed by a deep convolutional neural network (CNN) operating only on multiscale visual inputs and a lightweight text encoder constructed by multilayer perception (MLP) reconstructing spatiotemporal embeddings. Considering pretraining costs and model performance inspired by Radford et al. [21], we reconstruct the spatiotemporal text input with simplified prompt templates (R2STFT) to help

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addressing the heterogeneity in metadata files derived from multisource optical remote sensing. Subsequently, to decrease the storage costs and computational complexity, we still use a hash layer to transform the high-dimensional features into binary codes. Finally, a hierarchical semantic tree (RSHST) is proposed to improve the feature correlation between multisource optical remote sensing images, and new samples are generated via RSHST. In particular, these new correlation-aware sample pairs are reinput into the model to accelerate learning semantic correlations by contrastive loss, promoting the semantic relevance of index codes.

The main contributions of this article can be summarized as follows.

1) We design a typical cross-modal pretraining framework with an asymmetric dual-encoder architecture for extracting and fusing two modalities of features from multisource optical remote sensing images. In addition, a self-supervised contrastive loss is used before fusing and in the hash layer simultaneously, which enables more vision and text learning.
2) Considering the ambiguity and heterogeneity of multisource optical remote sensing metadata, we reconstruct the spatiotemporal text input with simplified prompt templates (R2STFT), improving model accessibility by lowering the entry bar of model execution. Subsequently, we devise an RSHST to facilitate multiple feature representation, reducing semantic-aware indexing structure feature sparsification.
3) Positive and negative sample pairs that are notably correlation aware for model pretraining are generated via RSHST. Compared with the mainstream end-to-end hash learning method and server-side retrieval engine, our experimental results show that CCLS2T performs better in typical remote sensing retrieval scenarios.

The remainder of this article is organized as follows. In Section II, we introduce related work from the aspect of remote sensing IR methods. Section III explains the proposed CCLSTE algorithm in detail, including its overall structure, image–text contrastive (ITC) learning, contrastive hash learning, RSHST, and data bootstrapping. Section IV presents the experimental results, including the experimental datasets, experimental settings, experimental results, ablation studies, and parameter analysis. Conclusion and future work are presented in Section V.

II. Related Works

This section provides an overview of the existing methods from three aspects: RS IR, vision-language pretraining, and vision-language pretraining in remote sensing.

A. RS Image Retrieval

Many studies have been dedicated to provide solutions for remote sensing IR (RSIR), including text-based, content-based, hashing-based, and server-side retrieval. TBIR involves searching for similar images through manually annotated labels and metadata [28], [29], [30], [31], [32], such as geographic location, acquisition time, or sensor type. For example, Qu et al. [30] combined the metadata characteristics of remote sensing data, devising a rapid access method based on XML for multisource satellite data. Wu and Fu [32] proposed metadata-based index storage management on a massive remote sensing image database. However, text-based retrieval not only consumes a significant amount of human effort in label annotation but also often produces inconsistent results, given that the visual content of images cannot be comprehensively conveyed through annotations and metadata. In contrast, content-based retrieval exploits the visual information of images to improve retrieval accuracy. This approach first extracts representative features of images and subsequently calculates the resemblance between the query and target images to search for the most analogous content. Initially, handcrafted features, including color features, texture features, and shape information, were commonly used [33], [34], [35]. For example, Taheri et al. [35] extracted low-level features and used deep Boltzmann machines to learn midlevel features and correlations between low- and mid-level features. Recently, benefiting from the formidable feature learning capacity of CNNs, many high-level features have been extracted by CNNs [9], [36], [37], [38], [39]. For example, Ge et al. [36] extracted the features of VGG16 and GoogLeNet under diverse input sizes to obtain multiscale information. Cao et al. [37] designed an RSIR approach using a triplet deep metric learning CNN, allowing the use of simple metric measures such as the Euclidean distance in direct image similarity comparisons. Shao et al. [38] introduced a multilabel RSIR approach using a fully convolutional network for region-based multilabel retrieval. However, content-based retrieval methods exhibit certain limitations. Low-/mid-level features underperform in RSIR due to the limited capacity for representing remote sensing images [40]. Moreover, despite their effectiveness, deep features hinder retrieval efficiency because of their high dimensionality [41], particularly in large-scale remote sensing images. In contrast, hashing-based retrieval considers retrieval speed and storage. This method encodes high-dimensional features into binary hash codes. Numerous hashing-based retrieval methods have been proposed, demonstrating excellent performance [42], [43], [44], [45], [46]. For instance, Song et al. [43] proposed a deep hashing CNN to obtain visually analogous images and provide semantic labeling concurrently. Liu et al. [44] proposed an RSIR method, feature and hash (FAH), incorporating a feature extraction model and an adversarial hash learning model. Tang et al. [46] developed a supervised hash-learning method on the basis of metalearning for large-scale RSIR tasks. In recent years, there has been an increase in contrastive hashing-based retrieval methods [47], [48], [49]. These approaches aim to boost retrieval accuracy by extracting distinct features, shielding against visual differences among multisource remote sensing images in the same categories. Nevertheless, hashing-based retrieval transforms high-dimensional data into a low-dimensional space, potentially resulting in information loss. In the context of remote sensing images, this approach can lead to the omission of crucial geographical details, such as object location and terrain characteristics.

In addition, the above retrieval methods both operate at the algorithmic rather than at the systematic level. These

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findings suggest that these methods have yet to be extensively applied in practical engineering applications. Consequently, numerous researchers have grappled with this challenge by developing server-side retrieval engines, which can be broadly categorized into two groups: storage-side retrieval and service-side retrieval. Storage-side retrieval builds multidimensional indexes within the database based on the storage structure of geospatial data. For example, SpatialLite [50] offers a KD tree and quad tree to optimize the querying and analysis of geospatial data. PostGIS [51], [52], [53] provides a KD tree, GiST, and SP-GiST to enhance the performance of the range query. MongoDB [54] supports the 2-D and geohash indexes and efficiently handles geolocation-related queries, such as nearby location searches and range queries. Service-side retrieval constructs a multidimensional indexing layer between the service and database layers, establishing mapping relations of geospatial data to multidimensional indexes. A notable example is GeoNode, a geospatial data management system built on GeoServer [55]. It employs grid-based indexing methods to improve service performance, leveraging the built-in GeoWebCache. GeoMesa [56], GeoTrellis [57], and GeoWave [58] are open-source geospatial data processing frameworks. These frameworks endorse grid-based indexing approaches, such as the Z-order and Hilbert indexes. In addition, they provide generalized indexing techniques, including quad tree, KD-tree, and R-tree indexing. These indexing methods can be customized to support diverse data distributions and fulfill specific query requirements.

B. Vision-Language Pretraining

Vision-language pretraining aims to establish a connection between vision and language through joint learning of (image, text) data or (video, text) data [59]. Self-supervision is used for pretraining, enabling the model to acquire universal representations and improving performance across various downstream tasks. There are two categories of vision-language pretraining in existing studies [55]. The first category focuses on developing separate unimodal encoders for images and text [60], [61], [62], [63], [64]. For instance, Harold Li et al. [60] proposed VisualBERT, which adopts a bidirectional encoder representations from transformers (BERT)-like text encoding format for text input and an image encoding format using a target detection algorithm for image input. Kim et al. [61] introduced vision-and-language transformer (ViLT), which employs a pretrained vision transformer (ViT) for visual feature processing. This process is coupled with text encoding to derive the ultimate encoding, which is subsequently fed into the transformer for further input. Li et al. [63] introduced Unicoder-VL, utilizing fast-RCNN for regional feature extraction, and generated regional object labels and regional spatial location features. These, together with text encoding, are sent to the transformer for pretraining. The other category concentrates on capturing the interactions between image features and text features by utilizing multimodal encoders based on transformers with a cross-attention mechanism [21], [65], [66], [67], [68]. A typical model is vision-and-language BERT (ViLBERT), introduced by Lu et al. [65]. In this model, a coattention transformer is used for cross-modal feature fusion. Bootstrapping language-image pre-training (BLIP) [68] is a multimodal hybrid encoder–decoder that can achieve single-modal encoding of images or text, image-based text encoding, and image-based text decoding.

Contrastive learning, as a self-supervised learning method, demonstrates remarkable cross-modal alignment and zero-shot learning capabilities in vision-language pretraining. For example, Radford et al. proposed [71] contrastive language-image pre-training (CLIP), which adopts the contrastive learning method to fetch and encode features in images and texts and subsequently calculates the cosine distance of (image, text) pairs. Thus, a connection between the two modalities is established. Similarly, momentum contrast (MoCo) uses contrastive learning to understand the correlation between images and text, with a specific emphasis on employing momentum contrast to enhance learning image features [23]. Yang et al. [66] proposed a triple contrastive learning (TCL) approach, making the acquired representations expressive in each unique modality and cross modalities. Zhang et al. [69] introduced a framework, 3D vision-language pre-training with object contrastive learning (3DVLP), to align objects with explanations and distinguish objects in a scene. Jiang et al. [70] employed a cross-modal contrastive learning strategy to optimize the mutual information between an image/text anchor and its negative texts/images by iteratively minimizing cross-modal similarity.

C. Vision-Language Pretraining in Remote Sensing

More recently, visual-language pretraining has gained widespread adoption in the remote sensing field to boost the presentation of downstream tasks, such as semantic segmentation [71], [72], scene classification [73], [74], [75], image–text retrieval [76], [77], [78], [79], and object detection [80], [81]. For instance, Chen and Bruzzone [71] explored semantic segmentation for remote sensing images utilizing a conditional U-Net model for predicting segmentation masks based on text descriptions and a pretrained CLIP model for aligning images and corresponding text. Zhang et al. [72] created a pipeline named text2seg, which utilizes multiple foundation models, such as GPT-4, to enhance the process of semantic segmentation in remote sensing images guided by a text prompt. Qiu et al. [73] leveraged a pretrained CLIP model used in feature extraction, achieving high-accuracy classification results even with limited samples. Li et al. [74] applied a pretrained vision language model for remote sensing scene classification based on the pretrained CLIP (RS-CLIP) model for semantic-aware visual representation and developed a pseudolabeling technique to address zero-shot remote sensing image classification. Yang et al. [76] employed a model in cross-modal remote sensing text-IR, capturing abundant semantic information and representations from visual modality. Yu et al. [77] proposed a cross-modal feature-matching network based on a graph neural network, which can mitigate the effects of information misalignment while effectively modeling the feature associations between two modalities. Tang et al. [78] proposed an interacting–enhancing feature transformer for remote sensing image–text retrieval. Sun et al. [80] built a general cross-modal multitask model framework called RingMo, which leverages the masked autoencoding method

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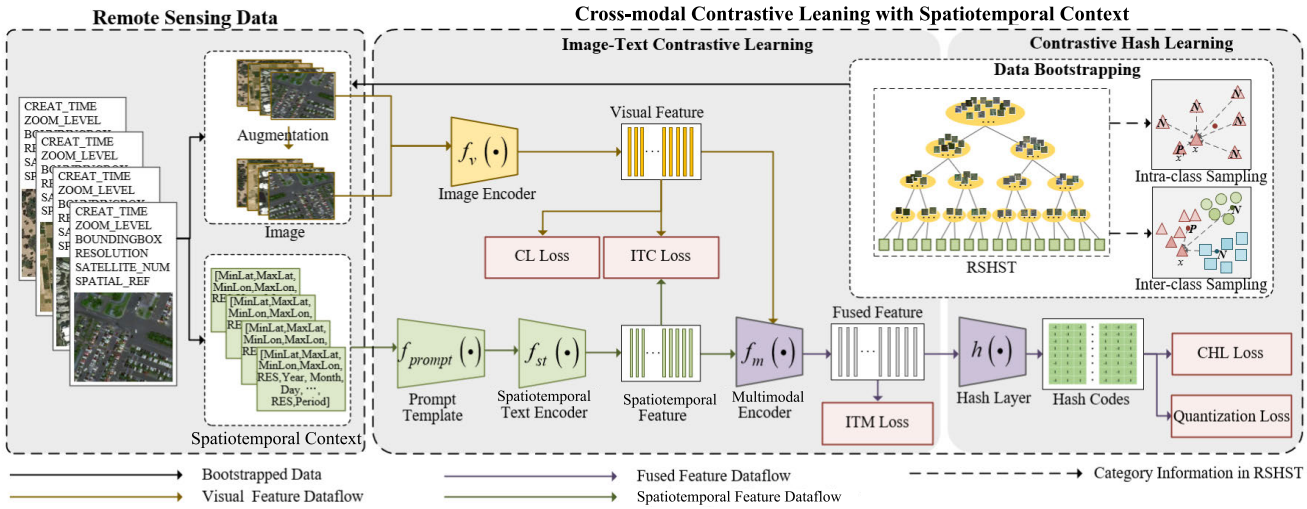


Fig. 2. Framework of our proposed CCLS2T.

and is adaptable to diverse tasks, for instance, object detection, scene classification, and semantic segmentation. Some studies also explore other downstream tasks, such as visual question answering [81], [82], [83] and image captioning [84], [85]. For example, Chappuis et al. [83] constructed a model for remote sensing visual question answering. Initially, context information is extracted through a visual model, which is subsequently transformed into text and integrated into a language model. Hu et al. [84] constructed a high-quality dataset for remote sensing image captioning (RSICap) and developed a foundation model based on generative pretrained transformers (GPTs), called the RSGPT.

III. Materials and Methods

The success of existing vision-language pretraining works, such as ViLT [52] and CLIP [54], has attracted widespread interest in the remote sensing community. Inspired by these authors, we designed a cross-modal pretraining model to explore the correspondence between images and metadata texts for correlation-aware MSRSIR. Fig. 2 shows the framework of CCLS2T. It contains an image encoder used for visual feature extraction, denoted by $f_v(\cdot)$, and a spatiotemporal text encoder used for spatiotemporal feature extraction, denoted by $f_{\mathrm{st}}(\cdot)$. We perform contrastive learning on these images to align the initial relationships between the visual and spatiotemporal features. The extracted features are fused through simple concatenation. Then, contrastive hash learning is performed, binarizing the fused features into compact hash codes, denoted by $h(\cdot)$. Moreover, in data bootstrapping, we carried out hierarchical K-means clustering via dynamic similarity division to construct an RSHST, which is used in intraclass sampling, interclass sampling, and hard negative sample mining (HNSM) for training pair optimization.

A. ITC Learning

As shown in Fig. 2, CCLS2T jointly trains an image encoder and a spatiotemporal text encoder to predict the correct (image, text) pairings. Given optical remote sensing sample data, (image, text) pairs, each consisting of a visual image and its STC from a metadata file, are obtained. In particular, an image–text contrastive loss is adopted to learn preferable unimodal representations of an (image, text) pair before fusion.

In comparison with models designed for other natural images, the proposed model demands more sophisticated visual feature extraction capabilities. Therefore, we use DenseNet [85] as the image encoder backbone. Unlike AlexNet [86], GoogLeNet [87], VGGNet [88], ResNet [89], and other state-of-the-art CNN models, DenseNet avoids stereotypical model optimization approaches, such as deepening the network layers and widening the network structures. DenseNet uses a dense block layer to maximize the flow of information, where each layer takes input from prior layers and transfers its corresponding feature map to subsequent layers. In this way, DenseNet allows the previous features to be efficiently passed to the subsequent feature, realizing automatic feature reuse and alleviating the vanishing gradient problem to a certain extent. Specifically, N remote sensing images $X = \{x_1, x_2, \ldots, x_N\}$ are input into the image encoder, and a set of high-dimensional features $\boldsymbol{F_v} = \{\boldsymbol{f}_{\boldsymbol{v},1}, \boldsymbol{f}_{\boldsymbol{v},2}, \ldots, \boldsymbol{f}_{\boldsymbol{v},N}\}$ is output as visual features corresponding to X. For any remote sensing image x_i, its extracted visual feature vector $\boldsymbol{f}_{\boldsymbol{v},i} \in R^{1\times 128}$ can be denoted as follows:

$$\boldsymbol{f}_{\boldsymbol{v},i} = f_v(x_i; \theta_v) \tag{1}$$

where f_v is denoted as the feature extraction operation of the image encoder and θ_v is a parameter.

The spatiotemporal text encoder is a function $\boldsymbol{f}_{st}(x_i : \theta_{st})$ used to embed spatial-aware attributes (e.g., minlon, minlat, maxlon, maxlat, and sres) and temporal-aware attributes (e.g., year, month, date, period, and tres) of remote sensing images to a vector of d dimensions, where d is set to 128. Before text encoding, one issue we should encounter is the rarity of datasets in which the spatiotemporal text paired with the remote sensing image is identical. However, the great differences between the metadata of remote sensing datasets derived from different remote sensing platforms cannot be ignored. For example, in an MTL file, in the metadata of a LandSAT 8 remote sensing image, "PRODUCT_UL_CORNER_LAT" denotes the maximum latitude of the image, and "SCENE_CENTER_TIME" denotes the center time at which the image was taken. In an ISPIRE file,

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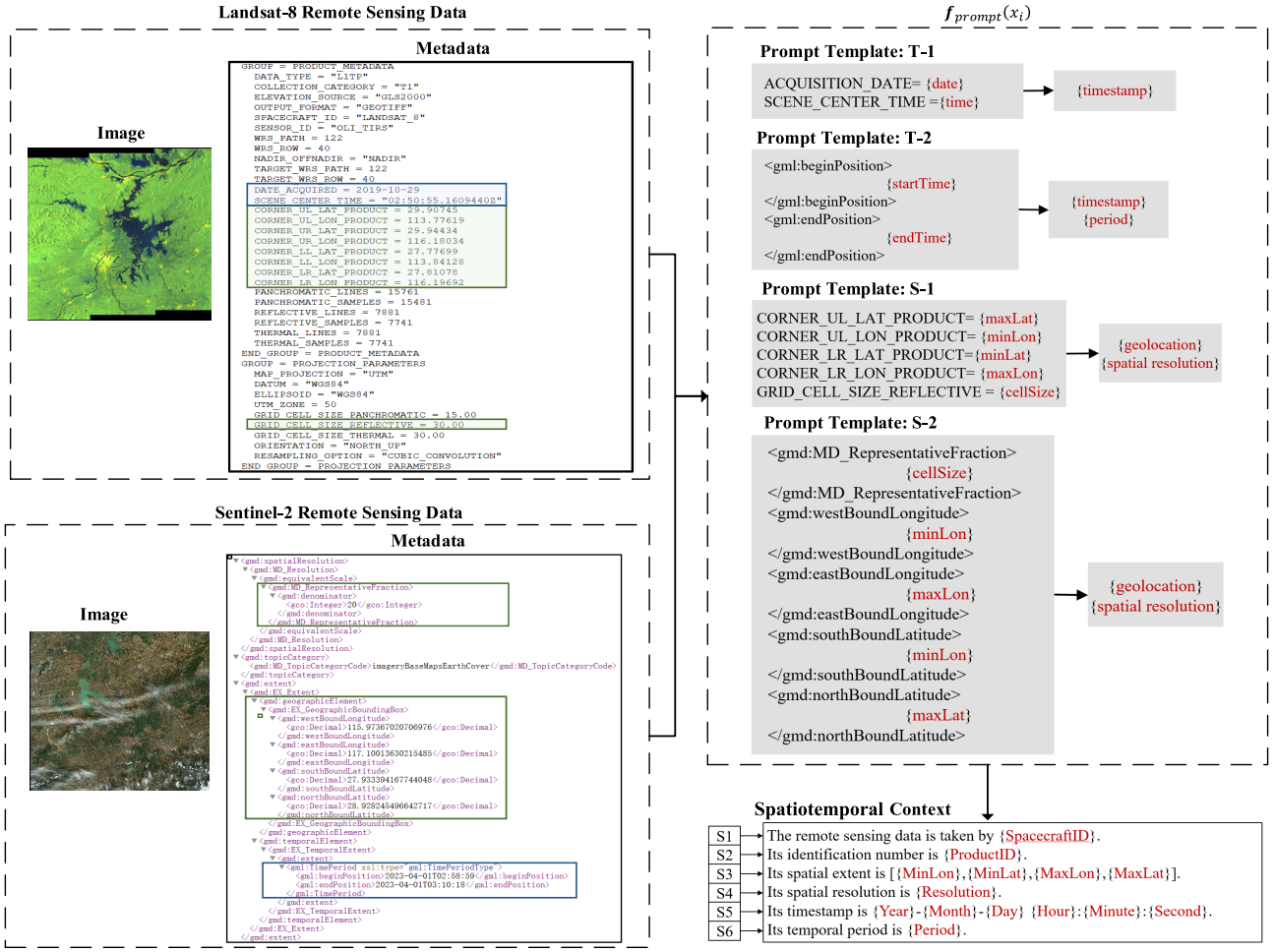


Fig. 3. Illustration of the proposed R2STFT.

in the metadata of a Sentinel-2 remote sensing image, "northboundLatitude" denotes the maximum latitude of the image. In addition, "beginPosition" and "endPosition" indicate the starting and ending times, respectively, of the image capture process. In addition, a large body of redundant information remain in metadata, interfering with the spatiotemporal attribute extraction. To help bridge this expression gap, similar to the "prompt engineering" discussion around GPT3 [90], [91], we design the spatiotemporal word filter (R2STFT), which uses simplified prompt templates to investigate spatial-aware and temporal-aware attributes from metadata files of remote sensing data. Transforming various spatiotemporal attributes from heterogeneous metadata into a unified textual space, which breaks the platform boundaries of multisource remote sensing data, is significant. The R2STFT algorithm represents a set of template-based rules for spatiotemporal attribute extraction and describes the unified structure extracted from the multisource remote sensing metadata, including structure, attributes, and values. Specifically, we define four prompt template rules to obtain spatiotemporal attributes from the raw data by mapping their metadata to suitable prompt templates, denoted by $f_{\text{prompt}}(x_i)$. As shown in Fig. 3, the spatiotemporal text contains not only the temporal and spatial attributes of the remote sensing data but also the data identifier and the acquisition device identifier. We use six tokens to identify these different contents and sequentially splice them into vectors of length 512, which are fed into the spatiotemporal text encoder for subsequent processing. Afterward, we use an MLP with an output size of 128 to build a spatiotemporal text encoder to ensure that the output spatiotemporal features are the same size as the visual features.

Moreover, since our framework focuses more on learning the hidden correlation features in multisource (image, text) pairs by vision-and-language pretraining with contrastive loss, a simple linear projection instead of cross attention [92] or dot product [21] is harnessed in a learned joint embedding space, which is denoted by $\boldsymbol{f}_m(x_i{:}\theta_m)$, to integrate the extracted features, expressed as $\boldsymbol{f} = \{\boldsymbol{f}_1, \boldsymbol{f}_2, \ldots, \boldsymbol{f}_N\}$. To explore the fine-grained semantic correspondence between visual and spatiotemporal features, we pretrain CCLS2T with three objectives: ITC learning and supervised classification learning (CL) on unimodal encoders and image–text matching (ITM) on a multimodal encoder.

ITC learning learns preferable unimodal feature representations before fusion with the loss function $\mathcal{L}_{\text{ITC}}$, denoted as follows:

$$\mathcal{L}_{\text{ITC}} = -\sum_{i=1}^{N} \log \frac{\exp\left(s\left(\boldsymbol{f}_{v,i}, \boldsymbol{f}_{st,i}\right)/\tau\right)}{\sum_{j=1}^{N} \exp\left(s\left(\boldsymbol{f}_{v,i}, \boldsymbol{f}_{st,j}\right)/\tau\right)} \quad \text{s.t. } i \neq j \tag{2}$$

where $s(\cdot,\cdot)$ denotes a similarity function [such as the cosine $(\cdot,\cdot)$].

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CL involves learning superior visual feature representations and improving the accuracy of semantic feature expression with a cross-entropy loss function $\mathcal{L}_{\mathrm{CL}}$ denoted as follows:

$$\mathcal{L}_{\mathrm{CL}} = -\sum_{k=0}^{C} y_k \log\left(p_k\left(\boldsymbol{f}_{\boldsymbol{v},i}\right)\right) \tag{3}$$

where $y_k \in \{\pm 1\}$ specifies the ground-truth class and $p_k(\boldsymbol{f}_{\boldsymbol{v},i}) \in [0, 1]$ is the output of a softmax function and indicates the probability of x_i being predicted as the kth class.

Moreover, we define an ITM loss $\mathcal{L}_{\mathrm{ITM}}$ to predict whether an (image, text) pair is matched. $\mathcal{L}_{\mathrm{ITM}}$ is denoted as follows:

$$\mathcal{L}_{\mathrm{ITM}} = -\sum_{i,j=1}^{N} y_{i,j} \log\left(p_{i,j}\left(\boldsymbol{f}_{\boldsymbol{v},i}, \boldsymbol{f}_{\boldsymbol{st},j}\right)\right) \tag{4}$$

where $y_k \in \{\pm 1\}$ represents the ground-truth label of ITM and $p_{i,j}(\boldsymbol{f}_{\boldsymbol{v},i}, \boldsymbol{f}_{\boldsymbol{st},j}) \in [0, 1]$ indicates whether the visual feature $\boldsymbol{f}_{\boldsymbol{v},i}$ matches the spatiotemporal feature $\boldsymbol{f}_{\boldsymbol{st},j}$.

Finally, the full objective of CCLS2T in the pretraining stage can be defined as follows:

$$\mathcal{L}_{\mathrm{DFE}} = \mathcal{L}_{\mathrm{CL}} + \lambda_{\mathrm{ITC}}\mathcal{L}_{\mathrm{ITC}} + \lambda_{\mathrm{ITM}}\mathcal{L}_{\mathrm{ITM}} \tag{5}$$

where λ_{ITC} and λ_{ITM} are used to balance the efforts of $\mathcal{L}_{\mathrm{CL}}$, $\mathcal{L}_{\mathrm{ITC}}$, and $\mathcal{L}_{\mathrm{ITM}}$.

B. Contrastive Hash Learning

We take contrastive hash learning as the downstream task to learn binarized representations of multiscale remote sensing data, whose output is used as remote sensing image indexes. Superior to the conventional hash learning method, contrastive hash learning can enhance the discriminative ability of hash index codes by improving their intraclass closeness and interclass variability, further improving the retrieval accuracy and efficiency. A fully connected layer with H nodes is used in contrastive hash learning, where H is the length of the hash code. We use the tanh$(\cdot)$ function as the activation function; therefore, the real-valued hash code can be expressed as follows:

$$\boldsymbol{h}_i = \tanh\left(\boldsymbol{f}_i; \theta_{\boldsymbol{H}}, \mathrm{bias}_{\boldsymbol{H}}\right) \tag{6}$$

where $\theta_{\boldsymbol{H}}$ and $\mathrm{bias}_{\boldsymbol{H}}$, respectively, denote the weights and the biases. In comparative hash learning, we use the inner product to calculate the cosine similarity of features; it is indispensable to normalize the real-valued hash code to obtain the normalized hash code, denoted as follows:

$$\boldsymbol{h}_{\boldsymbol{N},ih} = \frac{\boldsymbol{h}_{ih}}{\|\boldsymbol{h}_i\|_2} \tag{7}$$

where $h = 1, 2, \ldots, H$ denotes the hth element in the hash code of length H. Then, the normalized features $\boldsymbol{h}_{\boldsymbol{N},i}$ are input into the sign$(\cdot)$ function for binarization to obtain the hash code, denoted as follows:

$$\boldsymbol{b}_i = \mathrm{sign}\left(\boldsymbol{h}_{\boldsymbol{N},i}\right). \tag{8}$$

In conventional hash learning methods, the cross-entropy and quantization loss functions are usually used for defining learning objectives. Referring to [48], we replace the traditional cross-entropy loss function with the feature representation loss of contrastive learning, which is defined as follows:

$$\mathcal{L}_{\mathrm{CHL}} = -\sum_{i=1}^{N} \log \frac{\exp\left(\boldsymbol{h}_i \cdot \boldsymbol{h}_{j(i)}^{\mathrm{T}}/\tau\right)}{\sum_{x_r \in R(i)} \exp\left(\boldsymbol{h}_i \cdot \boldsymbol{h}_r^{\mathrm{T}}/\tau\right)} \tag{9}$$

where N is the count of the remote sensing images in a minibatch, τ is the temperature parameter, and $R(i)$ denotes a set of categories other than the category to which remote sensing image x_i belongs. Given is an anchor x_i, x_j represents a positive sample, while x_r represents a negative sample from the categories in $R(i)$. $\boldsymbol{h}_i$, $\boldsymbol{h}_{j(i)}$, and $\boldsymbol{h}_r$, respectively, denote the hash codes x_i, x_j, and x_r. We replace the hash code $\boldsymbol{h}_i$ with the real-valued code $\boldsymbol{h}_{\boldsymbol{N},i}$ to address the discrete optimization problem; then, $\mathcal{L}_{\mathrm{CHL}}$ is defined as follows:

$$\mathcal{L}_{\mathrm{CHL}} = -\sum_{i=1}^{N} \log \frac{\exp\left(\boldsymbol{h}_{\boldsymbol{N},i} \cdot \boldsymbol{h}_{\boldsymbol{N},j(i)}^{\mathrm{T}}/\tau\right)}{\sum_{x_r \in R(i)} \exp\left(\boldsymbol{h}_{\boldsymbol{N},i} \cdot \boldsymbol{h}_{\boldsymbol{N},r}^{\mathrm{T}}/\tau\right)}. \tag{10}$$

In addition, the quantization loss $\mathcal{L}_Q$ restricts the elements of $\boldsymbol{h}$ to approximately -1 or 1, bridging the gap between the real-valued and discrete hash codes [44]. Referring to [44] we define the quantization loss as follows:

$$\mathcal{L}_Q = -\frac{1}{N \times K} \sum_{i=1}^{N} \sum_{k=1}^{K} \left(\boldsymbol{h}_{ik} \log \boldsymbol{h}_{ih} + (1 - \boldsymbol{h}_{ih}) \log \boldsymbol{h}_{ih}\right) \tag{11}$$

where $\boldsymbol{h}_{ih}$ denotes the hth element in the hash code of x_i.

In summary, the loss function of contrastive hash learning $\mathcal{L}_H$ can be defined as follows:

$$\mathcal{L}_H = \mathcal{L}_{\mathrm{CHL}} + \lambda_Q \mathcal{L}_Q \tag{12}$$

where λ_Q is used to balance the efforts of $\mathcal{L}_{\mathrm{CHL}}$ and $\mathcal{L}_Q$.

C. Hierarchical Semantic Tree (RSHST)

The RSHST is a tree structure that can reflect the similarity of remote sensing images at different granularities. In RSHST, from the top to bottom layers, the remote sensing image categorization changes from coarse to fine, and the intraclass differences gradually decrease. We construct an RSHST using top–down remote sensing image division by clustering, which is illustrated in Fig. 4, where a darker color indicates a finer granularity of the division. The utilization of RSHST during data bootstrapping enables the generation of training pairs with variable differentiation properties, including intraclass sampling and interclass sampling, thus enhancing the ability of CCLS2T to extract discriminative features. We integrate K-means clustering into hierarchical clustering to classify remote sensing images hierarchically using feature distance in terms of the visual, temporal, and spatial dimensions for similarity judgment. We use visual distance to determine the remote sensing image division in the beginning. When it is difficult to distinguish them in the visual dimension further, we use spatiotemporal feature division instead of image division in the next layer, which is regarded as the final layer in the RSHST.

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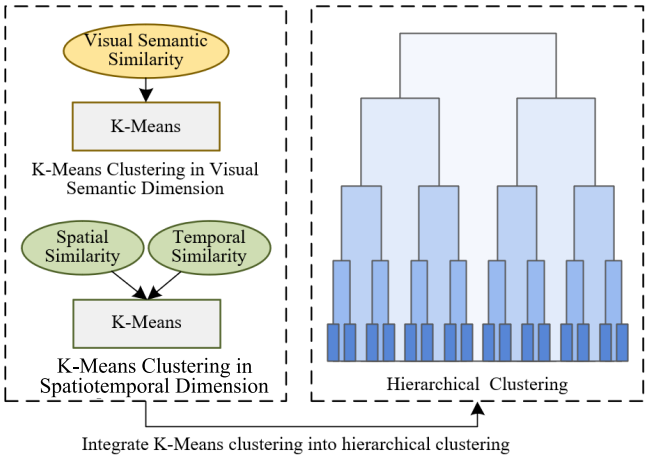


Fig. 4. Illustration of RSHST construction.

First, we regard all remote sensing images as belonging to a category in the origin layer, which can be called the origin category. Then, in the next layer, K-means clustering is performed on these remote sensing images. K elements are randomly selected as initial cluster centers and are continuously adjusted. We calculate the distance between each remote sensing image and each category center to determine which category the remote sensing image belongs to. K-means clustering is subsequently performed in the second layer on the remote sensing images in each category to divide them further into different categories. Similar clustering is repeated until the mean value of the visual distance between remote sensing images in each category is lower than a certain threshold ε_0, which indicates the end of the division of remote sensing images in the visual dimension. Clustering is then performed at the next layer in the temporal and spatial dimensions using temporal and spatial distances to determine the category to which each remote sensing image belongs. Thus, we finalize the remote sensing image division and construct an RSHST. At the beginning of clustering, we set the value of K to 2 by default. Then, we gradually increase the value of K until the mean feature distance of the remote sensing images in each category is less than the distance threshold ε_v (for visual clustering) or ε_{st} (for spatiotemporal clustering). The objective function of K-means clustering is

$$\min J = \sum_{i=1}^{M} \sum_{k=1}^{K} r_{i,k} D(x_i, u_k) \tag{13}$$

where u_k is the center of the category c_k, which can be expressed as the mean feature vector of all the remote sensing images in c_k. $r_{i,k}$ is 1 when x_i belongs to c_k; otherwise, $r_{i,k}$ is 0. $D(x_i, u_k)$ denotes the distance between the remote sensing image x_i and the category center u_k and can be expressed as follows:

$$D(x_i, u_k) = \begin{cases} D_v(x_i, u_k), & \overline{D_v} \geq \varepsilon_0 \\ \omega_t D_t(x_i, u_k) + \omega_s D_s(x_i, u_k), & \overline{D_v} < \varepsilon_0 \end{cases} \tag{14}$$

where $D_v(x_i, u_k)$, $D_t(x_i, u_k)$, and $D_s(x_i, u_k)$, respectively, denote the distances between x_i and u_k in terms of the visual dimension, temporal dimension, and spatial dimension. Algorithm 1 gives the corresponding pseudocode.

Algorithm 1 Construction of RSHST

Input: feature dataset $\boldsymbol{D}$, visual threshold regulator η_v, Spatiotemporal threshold regulator η_{st}, visual distance threshold ε_0.
Output: hierarchical semantic tree T;

```
Init T, Queue;
Put the data in D into Queue;
while (there are datasets in Queue for division)
    D_v = avg(Σ_{x_i,x_j∈D} x_i, x_j);
    ε_v = η_v D_v ;
    if ε_v > ε_0
      CV: K = 2;
      c = {select K points at random from D};
      while (the maximum number of iterations is not
            reached)
        for each remote sensing image x_i in D
          for each center c_k in c
            calculate D_v(x_i, u_k);
            if (D_v(x_i, u_k) = min(D_v(x_i, u)))
              D_k = D_k ∪ {x_i};
            end if
          end for
        end for
        for each center c_k in c
          D_v^c = avg(Σ_{x_i∈D_k} D_v(x_i, u_k));
          if (D_v^c > ε_v)
            K = K + 1;
            goto CV;
          end if
        end for
        for each center c_k in c
          T = T ∪ {D_k^c};
          Put D_k^c into Queue;
        end for
        break;
      end if
    else
      CST: K = 2;
      Select K points at random from D as c;
      while (the maximum number of iterations is not
            reached)
        for each data in D as x_i
          for each center in c as c_k
            Calculate D_st(x_i, u_k);
            if (D_st(x_i, u_k) = min(D_st(x_i, u)))
              D_k^c = D_k^c ∪ {x_i};
            end if
          end for
        end for
        for each center c_k in c
          D_st^c = avg(Σ_{x_i∈D_k} D_st(x_i, u_k));
          if (D_v^c > ε_v)
            K = K + 1;
            goto CST;
          end if
        end for
        T = T ∪ {D_k^c};
        break;
    end if
  return T;
```

Considering that there are large gaps in the data distribution between different datasets, it is necessary to dynamically

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adjust the distance threshold value accordingly. First, we calculate the average feature distance of the remote sensing images to be divided. Then, the distance thresholds ε_v and ε_{st} are dynamically set using the average feature distance multiplied by a weight coefficient. Thus, the visual distance threshold can be expressed as follows:

$$\varepsilon_v = \eta_v \overline{D_v} = \eta_v \frac{2}{n \times (n-1)} \sum_{i=1, j=1}^{n} D_v(x_i, x_j) \tag{15}$$

where $D_v(x_i, x_j)$ is the visual distance of remote sensing image x_i and x_j and $\overline{D_v}$ is the mean visual distance of dataset X containing n elements. When the average value of the distance between the remote sensing image and the category center is less than ε_v, we complete the current iteration and use the results as the final output of the remote sensing image division at this layer. After the clustering at the first layer is completed, the segmentation of the remote sensing image belonging to each category will continue at the second layer. This iterative process persists until $\overline{D_v}$ of the remote sensing images in each category are currently below ε_0, a certain threshold in the visual dimension, indicating that the remote sensing images are extremely close in terms of visual semantics in the category with a high degree of visual similarity. Therefore, further segmentation in the visual dimension is stopped when the average distance between the remote sensing images in a category is less than ε_0. Then, we perform spatiotemporal feature clustering at the next layer using the spatiotemporal proximity of the remote sensing images. Like in ε_v, the spatiotemporal distance threshold ε_{st} can be expressed as follows:

$$\begin{aligned} \varepsilon_{st} &= \eta_{st} \overline{D_{st}} = \eta_{st} \frac{2}{n \times (n-1)} \sum_{i=1}^{n} \sum_{j=1}^{n} D_{st}(x_i, x_j) \\ \text{s.t. } D_{st}(x_i, x_j) &= \omega_t D_t(x_i, x_j) + \omega_s D_s(x_i, x_j), \quad i \neq j \end{aligned} \tag{16}$$

where $D_t(x_i, x_j)$ and $D_s(x_i, x_j)$ are the temporal and spatial distances of x_i and x_j, respectively, and ω_t and ω_s are used to balance their efforts.

Subsequently, the distances between any remote sensing image in terms of the visual, temporal, and spatial are introduced in detail.

1) Visual Distance: Remote sensing images always have a certain degree of similarity in the visual dimension when describing targets with the same semantics. We can extract visual features from remote sensing images and calculate the distance between them in a high-dimensional space to measure such similarities. Thus, the visual semantic distance is expressed as follows:

$$D_v(x_i, x_j) = \left\| \boldsymbol{f}_{v,i} - \boldsymbol{f}_{v,j} \right\| \tag{17}$$

where $\|\boldsymbol{f}_{v,i} - \boldsymbol{f}_{v,j}\|$ represents the distance between the visual feature vectors of x_i and x_j.

2) Temporal Distance: Temporal distance reflects the similarity between remote sensing images in the temporal dimension. Considering whether remote sensing images exhibit periodic changes in the temporal dimension, remote sensing images can be divided into two groups: periodic remote sensing images and nonperiodic remote sensing images. For periodic remote sensing images, time distance is denoted using the relative differences between the temporal attributes within a period, while for nonperiodic remote sensing images, time distance is denoted using the absolute differences between the temporal attributes of remote sensing images. The expression of the temporal distance is as follows:

$$D_t(x_i, x_j) = \operatorname{mod}\left(\left|t_i - t_j\right|, T\right) \tag{18}$$

where t_i and t_j denote the shooting times of x_i and x_j in milliseconds, respectively, and $|t_i - t_j|$ denotes their difference. T is the length of the period of x_i and x_j. Specifically, T is ∞ for remote sensing images.

3) Spatial Distance: Spatial distance denotes the similarity between remote sensing images in the spatial dimension and can be evaluated from the spatial resolution, topology, and location perspectives. We define spatial distance as the product of topological distance and location distance, expressed as follows:

$$D_s(x_i, x_j) = D_{TP}(x_i, x_j) \cdot D_{\mathrm{DIS}}(x_i, x_j) \tag{19}$$

where $D_{\mathrm{TP}}(x_i, x_j)$ and $D_{\mathrm{DIS}}(x_i, x_j)$, respectively, denote the topological distance and location distance between x_i and x_j.

The topological distance is used to measure the topological correlation between remote sensing images with six defined topological correlations: disjoint, contact, overlap, containment, inclusion, and equality. The topological distance between uncorrelated remote sensing images is maximized; the topological distance between remote sensing images with equal topological correlation is minimized. The topological distances corresponding to other correlations take values in the range of 0–1.

In addition, the location distance is used to express the spatial distance between the two remote sensing images in quantity, where the difference between their spatial grid codes is utilized.

D. Data Bootstrapping

To accelerate CCLS2T training and improve its retrieval accuracy, it is critical to increase the training dataset quality and select suitable positive and negative samples with balanced distributions are critical concerns. Therefore, we carry out data bootstrapping from the following perspectives: image augmentation (IA), intraclass sampling, interclass sampling, and HNSM.

1) Image Augmentation: Although the data acquisition capability of remote sensing methods is powerful, problems remain in applying remote sensing images to training models, such as the high cost of manual labeling. Thus, deep learning models often suffer from overfitting, leading to poor testing accuracy and generalization ability, making it difficult to meet the application requirements [91]. This problem can be solved by IA to generate new images and enrich training datasets. When performing image enhancement, generating as many images as possible is necessary in a way that does not change the original semantics, relying on in-depth thinking about the nature of remote sensing images. Considering the

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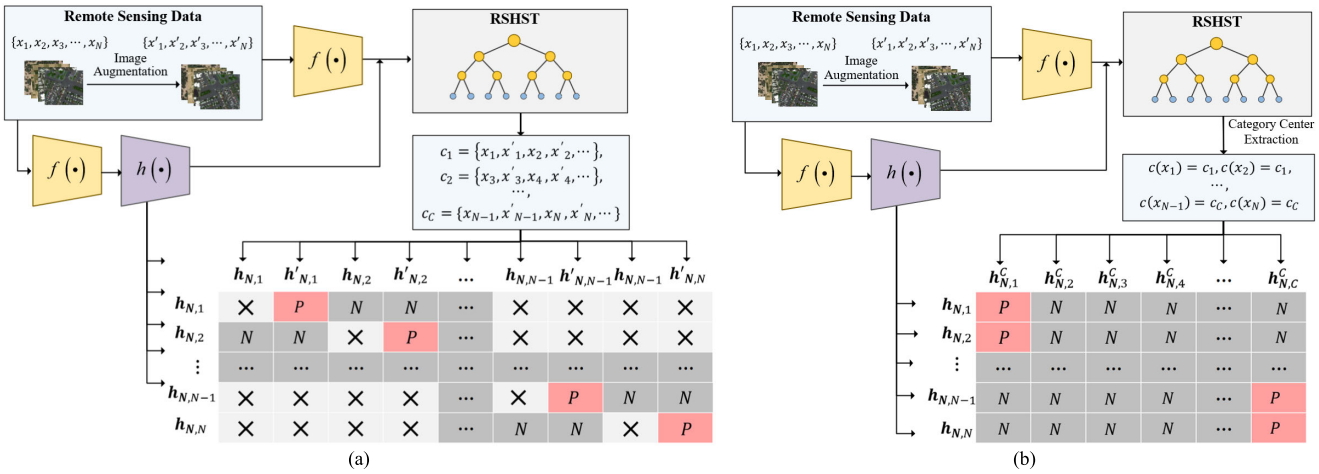


Fig. 5. Illustration of intraclass sampling and interclass sampling based on RSHST. (a) Intraclass sampling. Given any remote sensing data, take its image enhancement result is a positive sample, and take other remote sensing data of the same class as negative samples. (b) Interclass sampling. Given any remote sensing data, take the category center of the class it belongs to as a positive sample, and take the category centers of other classes as negative samples.

vertical shooting mode of remote sensing images and internal characteristics, such as multiple scales, multiple features, and cloud occlusion, we adopt the following strategies for IA.

1) We rotate each image by 90°, 180°, and 270°.
2) The stochastic local erasure method is employed when simulating occlusion in optical remote sensing images. There exists a distinction between optical remote sensing images and the images produced by other remote sensing techniques, which necessitates the consideration of cloud and atmospheric occlusion. Consequently, the random local erasure approach is employed to remove the image information located in the local areas of the remote sensing images. This process entails the random placement of opaque regions with various sizes and locations on each remote sensing image.
3) Given the multiple semantics in remote sensing images, we apply the hybrid method of random cropping and stitching, seamlessly combining remote sensing images in close spatial proximity to generate new images.
4) We utilize the spatial pyramid technique to simulate the feature heterogeneity illustrated in Fig. 1(b). For any remote sensing image, we resample it to obtain new images at different spatial scales and slice them sequentially into tiles of the same size. We take these tiles to supplement the remote sensing image datasets used for multiscale contrastive learning.

2) Intraclass Sampling: Intraclass sampling involves training pairs of remote sensing data in the same categories. It is used to differentiate between intraclass remote sensing data with strong similarities. In intraclass sampling, a data transformation operation is adopted for images in the minibatch $X_{(N)}$ by applying the above IA operation to each image x_i and combining the results with the original spatiotemporal features of x_i. Thus, the minibatch is extended to twice its original size, denoted as $X_{(2N)} = \{x_1, x_1', x_2, x_2', \ldots, x_N, x_N'\}$. Then, the similarity analysis is applied to the minibatch $X_{(N)}$ to construct RSHST. $C = \{c_1, c_2, \ldots, c_K\}$ indicates K categories corresponding to leaf nodes in RSHST, $c(x_i)$ is the category to which x_i belongs, and $N(c(x_i))$ denotes the count of images in $c(x_i)$. For any image x_i in $X_{(N)}$, we take its transformation result x_i' with IA as a positive sample to construct a positive pair and take remote sensing images belonging to $c(x_i)$ except x_i and x_i' as negative samples to form $N(c(x_i)) - 2$ negative pairs.

To summarize, intraclass positive pairs $P_{IC} = \{(\boldsymbol{h}_{N,i}, \boldsymbol{h}'_{N,i})\}$ denote the pairs containing remote sensing data x_i and transformation results x_i', as marked with red boxes in Fig. 5(a); intraclass negative pairs $N_{IC} = \bigcup_i N_{IC,i}$ indicate all mismatching remote sensing data pairs in $X_{(2N)}$, marked with gray boxes, where $N_{IC,i} = \{(\boldsymbol{h}_{N,i}, \boldsymbol{h}_{N,r}) | x_r \in c(x_i)/\{x_i, x_i'\}\}$ indicates $N(c(x_i)) - 2$ negative pairs for x_i.

3) Interclass Sampling: Interclass sampling involves training pairs of remote sensing data in different RSHST categories and is used to distinguish cross-category remote sensing data with strong similarity. In interclass sampling, positive and negative samples are chosen based on the constructed RSHST, similar to the intraclass sampling. For any remote sensing data x_i in $X_{(N)}$, the center of the category $c_j = c(x_i)$ is regarded as its corresponding positive sample, and the centers of other categories in $C = \{c_1, c_2, \ldots, c_K\}$ are regarded as corresponding negative samples, in which c_j is excluded.

In summary, interclass positive pairs $P_{\text{EC}} = \{(\boldsymbol{h}_{N,i}, \boldsymbol{h}^C_{N,j})\}$ denote the pairs containing x_i and the center of the category $c_j = c(x_i)$, as marked with red boxes in Fig. 5(b). RSHST-based interclass negative pairs $N_{\text{EC}} = \bigcup_i N_{\text{EC},i}$ indicate all negative pairs formed on $X_{(2N)}$, marked with gray boxes, where $N_{\text{EC},i} = \{(\boldsymbol{h}_{N,i}, \boldsymbol{h}^C_{N,r}) | c_r \in \{c_1, c_2, \ldots, c_K\}/\{c(x_i)\}\}$ indicates $K - 1$ negative pairs for x_i.

Considering the intraclass sampling and interclass sampling we adopted, the contrastive learning loss $\mathcal{L}_{\text{CHL}}$ comprises two corresponding components, expressed as follows:

$$\mathcal{L}_{\text{CHL}} = \mathcal{L}^I_{\text{CHL}} + \lambda_E \mathcal{L}^E_{\text{CHL}} \tag{20}$$

where $\mathcal{L}^I_{\text{CHL}}$ denotes the contrastive learning loss corresponding to the training pairs generated by intraclass sampling and $\mathcal{L}^E_{\text{CHL}}$ refers to the contrastive loss corresponding to the training pairs generated by interclass sampling. λ_E is used to balance them. Specifically, $\mathcal{L}^I_{\text{CHL}}$ can be defined as

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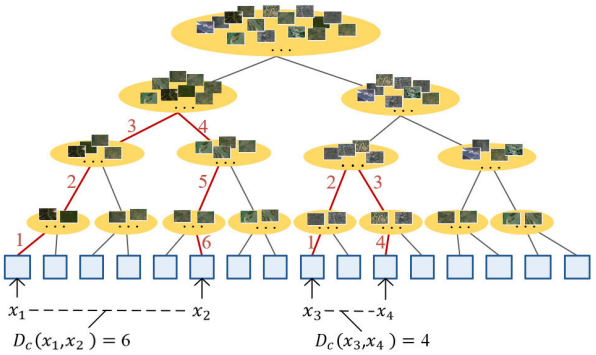


Fig. 6. Category distances between remote sensing data in RSHST.

follows:

$$\mathcal{L}_{\mathrm{CHL}}^{I}=-\sum_{i=1}^{N}\log\frac{\exp\left(\boldsymbol{h}_{N,i}\cdot\boldsymbol{h}'^{\mathrm{T}}_{N,i}\Big/\tau\right)}{\sum_{x_r\in R_I(i)}\gamma(x_i,x_r)\exp\left(\boldsymbol{h}_{N,i}\cdot\frac{\boldsymbol{h}^{\mathrm{T}}_{N,r}}{\tau}\right)} \tag{21}$$

where $\boldsymbol{h}'_{N,i}$ denotes the feature of x'_i, the positive sample of x_i is transformed from x_i utilizing IA, and $\boldsymbol{h}_{N,r}$ denotes the feature of x^r in $R_I(i)$, a set of negative samples if x_i consists of elements in c_j except x_i and x'_i. $c_j=c(x_i)$ is the category in which x_i belongs. $\mathcal{L}_{\mathrm{CHL}}^{E}$ can be defined as follows:

$$\mathcal{L}_{\mathrm{CHL}}^{E}=-\sum_{i=1}^{N}\log\frac{\exp\left(\boldsymbol{h}_{N,i}\cdot\boldsymbol{h}^{C\,\mathrm{T}}_{N,j}\Big/\tau\right)}{\sum_{c_r\in R_E(i)}\gamma(x_i,c_r)\exp\left(\boldsymbol{h}_{N,i}\cdot\boldsymbol{h}^{C\,\mathrm{T}}_{N,r}\Big/\tau\right)} \tag{22}$$

where $\boldsymbol{h}^{C}_{N,j}$ denotes the feature of the category center c_j to which x_i belongs, and $\boldsymbol{h}^{C}_{N,r}$ denotes the feature of a category center c_r in $R_E(i)$, a set of category centers corresponding to the leaf nodes in RSHST except c_j.

4) Hard Negative Sample Mining: Considering the reality that there are large negative samples and a few positive samples in a dataset, this approach may lead to a trend toward treating the data as negative samples. Therefore, the dataset needs to be trimmed to maintain stability between positive and negative samples. In addition, confusing remote sensing data tend to have representations that are similar to each other. Therefore, we should focus more on these hard negative samples, especially those with indistinguishable representations.

We propose a HNSM method that relies on the constructed RSHST model to address the aforementioned problems. We define the category distance in RSHST, which is utilized to balance the contributions of different negative samples to the loss function. Conversely, negative samples with shorter category distances to the anchor have greater impacts, allowing CCLS2T to focus on the nuances of negative samples with high similarity. We define the category distance of any two remote sensing images as the number of hops between the leaf nodes in RSHST corresponding to the categories to which they belong, as shown in Fig. 6. In Fig. 6, $c_1, c_2, \ldots, c_{16}$ and represent categories on the leaf nodes in RSHST; x_i, x_j, x_m, and x_n, respectively, denote any remote sensing image in categories c_1, c_5, c_9, and c_{12}; $D_c(x_i, x_j)$ is the category distance between x_i and x_j; and $D_c(x_m, x_n)$ is the category distance between x_m and x_n. We highlight the paths in the category distance computation with red lines and label the values corresponding to each hop. As shown in Fig. 6, remote sensing images in different semantic categories have greater category distances than do those in the same semantic categories but different spatiotemporal categories. $\gamma(x_i, x_r)$ is used to regulate the contribution of the negative sample x_r to the loss function, defined as follows:

$$\gamma(x_i,x_r)=\beta^{\ln(D_c(x_i,x_r)-1)} \tag{23}$$

where x_i and x_r are a negative pair generated by the above interclass sampling and $D_c(x_i, x_r)$ is the category distance between x_i and x_r. β is an attenuation factor that enables $\gamma(x_i, x_r)$ to decrease rapidly with increasing category distance. According to expert experience, we set β to 0.05 and select negative pairs satisfying $\gamma(x_i, x_r) > 0.1$ for the calculation, indicating that $D_c(x_i, x_r) \leq 6$.

IV. Experimental Results and Analysis

In this study, we perform experiments on three published optical remote sensing image datasets to evaluate the retrieval effectiveness and performance of our proposed CCLS2T model for the MSRSIR task.

A. Datasets and Evaluation Metrics

The EuroSAT dataset [96] is a benchmark for deep learning in land use and land cover classification. The dataset comprises Sentinel-2 satellite images and contains 27 000 images distributed among ten categories. Each category has 2000–3000 optical remote sensing images, all of which are 64 × 64 pixels in size.

The Million-AID dataset [97] has 51 categories, where the number of optical images ranges from approximately 2000–45 000, giving the dataset a long-tailed distribution. In addition, the Million-AID dataset has strengths, such as high spatial resolution, large scale, and global distribution and is superior to existing scene classification datasets. In contrast to other datasets, the Million-AID dataset exhibits noticeable feature heterogeneity within categories and feature homogeneity between categories.

The LivingAtlas dataset contains optical images of 12 different categories that were selected from the ArcGIS Living Atlas Platform [98], such as forests, farmlands, meadows, residential areas, rivers, and salterns. For each class, 1000 optical images measuring 256 × 256 pixels were manually selected at different times. Each optical image in this dataset has category annotations, temporal, and spatial attributes. We obtain the LivingAtlas dataset through the following operations. We identified the locations of the objects belonging to different categories on the ArcGIS Living Atlas Platform and obtained images corresponding to these locations on the platform. Then, we convert the locations to the level, row, and column order in the tile pyramid, combined with the version to build requests for optical remote sensing images in JPG format. We manually label each optical remote sensing image with its categorical information and take the release time on the ArcGIS Living Atlas Platform as the temporal attribute. We take the spatial extent and spatial resolution as spatial attributes, which can

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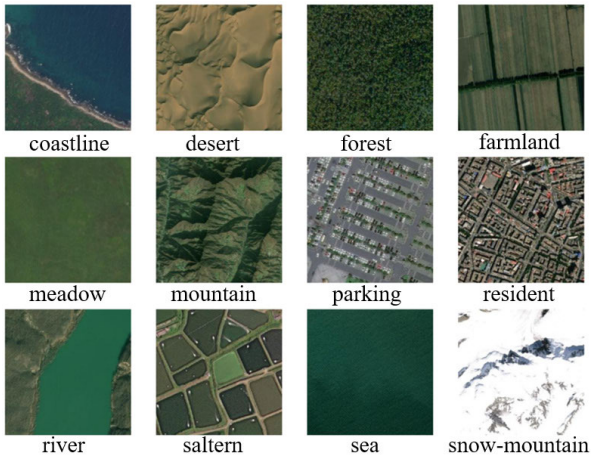


Fig. 7. Example images of classes in the LivingAltas dataset.

be computed using the level, row, and column order of the optical remote sensing image. Corresponding example images are given in Fig. 7. The original intention when constructing the Living Atlas dataset is for it to serve as a supplement to other datasets, enabling it to support performance comparisons conducted on both distinguishable and indistinguishable datasets. Therefore, when selecting data from the ArcGIS Living Atlas Platform to construct the Living Atlas dataset, we aim to minimize the differences between data belonging to the same class and maximize the differences between data belonging to different classes.

We adopt several widespread metrics to evaluate the accuracy and efficiency of the proposed CCLS2T model, including precision@k, recall@k, precision–recall, average precision (AP), average response time (ART), and the Davies–Bouldin index (DBI) [99].

The precision@k, recall@k, and precision–recall metrics are used to compare specific retrieval results in the visual dimension for more detailed results. The AP is used to evaluate the overall accuracy of RSIR and can be obtained by calculating the proportion of the remote sensing image satisfying the query conditions in the retrieval results to the total remote sensing images in the retrieval results, expressed as follows:

$$\mathrm{AP} = \frac{1}{|Q|}\sum_{i=1}^{|Q|}\mathrm{precision}(i). \tag{24}$$

ART denotes the total time spent from sending a retrieval request to receiving the retrieval results and consists of several components, such as the network transmission time and the retrieval time. In our experiments, we retrieve different remote sensing images consecutively over a period and calculate the average of the response times of all retrievals to obtain the ART.

The DBI is used as a metric for evaluating category delineation in the RSHST. It is computed by the tightness within the categories and the separation between the categories. The DBI can be expressed as follows:

$$\mathrm{DBI} = \frac{1}{k}\sum_{i=1}^{k}\max_{i\neq j}\frac{s_i + s_j}{d_{ij}} \tag{25}$$

where k is the number of categories, s_i denotes the average distance between the category center and all the remote sensing data in the ith category, and d_{ij} is the distance between the center of the ith and jth categories. $s_i + s_j \geq d_{ij}$ means that the category division is not suitable; otherwise, $s_i + s_j \ll d_{ij}$ means that the categories are well separated. A DBI value converging to 0 indicates a prime category delineation.

B. Experimental Settings

1) Retrieval Tasks: First, we apply the proposed CCLS2T algorithm to two widely explored types of remote sensing image search scenarios to validate its effectiveness in geospatial applications.

Range retrieval (RR) aims to locate remote sensing images within a multidimensional range; this approach is a common query scenario in the remote sensing domain. The typical query conditions for RR include the query image, spatial range, and temporal range and can be expressed as (image, sRange, tRange). The spatial range sRange can be denoted as sRange(xMin, yMin, xMax, yMax), where xMin, yMin, xMax, and yMax denote the minimum longitude, the minimum latitude, the maximum longitude, and the maximum latitude of the query window in the spatial dimension, respectively. The temporal range tRange can be denoted as tRange(tMin, tMax), where tMin and tMax denote the start time and stop time, respectively, of the query window in the time dimension. When performing the RR, we regard the query conditions in different dimensions as restriction parameters and input them into CCLS2T to obtain the outputs as the index boundaries. Then, we scanned and extracted remote sensing images within the index boundaries. Finally, we recheck and filter these remote sensing images to ensure that the returned results satisfy the multidimensional query conditions.

K-nearest neighbor (KNN) retrieval obtains the nearest K remote sensing images based on sorting by similarities between the remote sensing images and the query image. We input the query image into CCLS2T and take the output as the target index. Then, we start with the target index and scan $2K$ indexes forward and backward, respectively. Next, we calculate the Hamming distances to the target indexes to determine the similarity between the corresponding remote sensing images and the query image. Finally, we rank these remote sensing images and return the top K images with the greatest resemblance to the KNN results.

2) Implementation Details: To implement pretraining of our framework, some preparatory work should be completed. For the visual encoder, we treat the 121-layer model, DenseNet-121 [85], as the basic pipeline in our framework. We append a global average pooling layer and a fully connected layer to obtain the image vector and a batch normalization layer to regularize the vectors later. We take raw remote sensing images as the inputs of the visual encoder after resizing them to 64 × 64 × 3 pixels and apply the IA proposed in Section III-D1. We use only an MLP with two hidden layers for the spatiotemporal text encoder for word embedding. We apply the proposed R2STFT algorithm to the metadata files of remote sensing images to obtain STCs and take them as the input of the spatiotemporal text encoder. The

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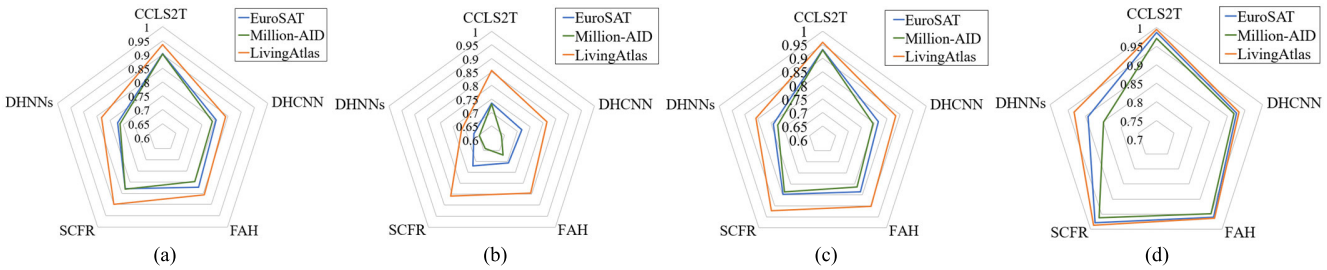


Fig. 8. Comparison of retrieval performance on different datasets with different metrics. (a) AP. (b) R@10. (c) R@50. (d) R@100.

TABLE I
CONFIGURATION OF PRETRAINED NETWORK USED IN ITC LEARNING

Encoder	Layer	Output Shape
Image Encoder	Input	64×64×3
	DenseNet121	Multiple
	Global Average Pool	1024
	Full (1)	128
	Batch Normalization	128
Spatiotemporal Text Encoder	Input	512
	Full (2)	1024
	Dropout	128
	Full (3)	128
Multimodal Encoder	Linear Projection	256

outputs of the spatiotemporal text encoder are taken as 128-D spatiotemporal features, the same size as visual features. For a multimodal encoder, a shallow operation (linear projection) is still used to imply the interaction between the image vector and the text vector we obtained. The corresponding architecture is given in Table I.

In training the proposed CCLS2T model, we set the mini-batch size to 128, the learning rate t to 0.0001, and the number of epochs T to 200. By default, we set $\eta_v = 0.07$, $\eta_{st} = 0.07$, $\lambda_{\text{ITC}} = 0.4$, $\lambda_{\text{ITM}} = 0.5$, $\lambda_E = 0.6$, $\lambda_Q = 0.6$, $\beta = 0.05$, and $H = 256$ when experimenting on three datasets. The influence of crucial hyperparameters is discussed in Section IV-E. Our experimental environment is the deep learning framework TensorFlow-GPU 2.4.3 and the operating system CentOS 7.*. We developed a remote sensing image access service that realizes RSIR by invoking CCLS2T in the background to facilitate the measurement of retrieval efficiency. We use Gatling [100] as the performance testing tool, which sends retrieval requests continuously, to compare the retrieval efficiency of the proposed CCLS2T and server-side retrieval engines.

C. Experimental Results

1) Comparison of Accuracy With Other State-of-the-Art Methods: To verify the efficacy of our method, we contrast CCLS2T with state-of-the-art methods on three optical datasets. Compared methods include deep hashing neural networks for large-scale RSIR (DHNNs) [42], deep hashing CNNs that retrieve images and classify them at once (DHCNN) [43], FAHlearning [44] and supervised contrastive hash learning using global and local features (SCFRs) [48].

Table II shows the top ten retrieved results on the three datasets. For each query example, the five different rows on the right are the results of DHNNs, DHCNN, FAH, SCFR, and CCLS2T. We mark the correct results with green rectangles and the incorrect results with red rectangles. The results presented in Table II demonstrate that the proposed CCLS2T method outperforms the other methods, particularly on the Million-AID dataset, whose images are challenging to distinguish due to heterogeneity within categories and feature homogeneity between categories.

Then, we perform a quantitative comparison of different RSIR approaches on the three datasets. Table III shows the results for the ART, AP, recall@10, recall@50, and recall @100 metrics, denoted by ART, AP, R@10, R@50, and R@100, respectively. According to the results, the CCLS2T algorithm proposed in this article outperforms other state-of-the-art methods in terms of retrieval accuracy and response time. Although the accuracy and recall of CCLS2T are lower than those of the SCFR method in some cases, there are tiny differences between them. For instance, the R@50 value of CCLS2T is only 0.0028 lower than the value of SCFH in the LivingAtlas dataset. Fig. 8 presents a comparative analysis among the results obtained on the three datasets by the four different methods, as quantified by four metrics denoted as AP, R@10, R@50, and R@100. As illustrated in Table III and Fig. 8, our proposed CCLS2T approach and the other four methods demonstrate superior performance on the Living Atlas dataset than on the other two datasets because of the characteristics of the Living Atlas dataset. The original intention when constructing the Living Atlas dataset is for it to serve as a supplement to other datasets, enabling it to support performance comparisons on both distinguishable and indistinguishable datasets. Therefore, when selecting data from the ArcGIS Living Atlas Platform to construct the Living Atlas dataset, we aim to minimize the differences between data belonging to the same class and maximize the differences between data belonging to different classes. This approach results in enhanced intraclass similarity and interclass separation, thus facilitating the extraction of discriminative features. The results indicate that SCFR exhibits some adaptability when extracting the features of optical remote sensing images with intraclass similarity and interclass dissimilarity. Nevertheless, the performance of SCFR is inferior to that of our proposed CCLS2T approach in terms of the R@50 and R@100 metrics due to the fact that SCFR is designed to focus excessively on typical visual features and is less capable of extracting fine-grained features from optical remote sensing images.

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TABLE II

RETRIEVAL EXAMPLES WITH TOP TEN RETRIEVED REMOTE SENSING IMAGES ON THREE DATASETS

Query	Results									
 river (Eurosat)	DHNNs	river	pasture	pasture	annualcrop	river	river	river	river	river	river
	DHCNN	river	river	forest	river	river	river	river	river	river	forest
	FAH	river	river	river	river	river	river	river	pasture	pasture	river
	SCFR	river	river	river	river	river	river	forest	river	river	river
	CCLS2T (ours)	river	river	river	river	river	river	river	river	river	river
 river (Million-AID)	DHNNs	river	river	river	river	river	forest	meadow	river	river	river
	DHCNN	river	river	river	river	farmland	farmland	river	river	river	meadow
	FAH	river	river	river	river	farmland	river	river	river	river	river
	SCFR	meadow	river	river	river	river	river	river	river	river	river
	CCLS2T (ours)	river	river	river	river	river	river	river	river	river	river
 river (LivingAtlas)	DHNNs	river	river	farmland	saltern	sea	river	river	river	river	river
	DHCNN	river	river	sea	river	river	river	river	river	coastline	river
	FAH	river	river	river	river	parking	river	river	river	river	river
	SCFR	river	mountain	river	river	river	river	river	river	river	river
	CCLS2T (ours)	river	river	river	river	river	river	river	river	river	river

In comparison, although CCLS2T exhibits slight inferiority in terms of the R@10 metric on the Living Atlas dataset, it outperforms the SCFR method with respect to the AP, R@50, and R@100 metrics, in addition to multiple metrics on other datasets. In general, CCLS2T demonstrates an absolute retrieval efficiency advantage in cases with similar retrieval accuracies. This result stems from the fact that our method carries out cross-modal contrastive learning on multisource optical remote sensing images and text, thus aligning their visual and spatiotemporal semantics.

Subsequently, we compare CCLS2T with other methods on the recall@k, precision@k, and precision–recall metrics. The corresponding results for the three datasets are given in Figs. 9–11. The results demonstrate that DHNNs and DHCNN

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TABLE III
COMPARISON OF CCLS2T WITH STATE-OF-THE-ART METHODS

Methods	ART (ms)	EuroSAT				Million-AID				LivingAtlas			
		AP	R@10	R@50	R@100	AP	R@10	R@50	R@100	AP	R@10	R@50	R@100
DHNNs	233	0.7698	0.6705	0.7905	0.8935	0.7623	0.6487	0.7742	0.8485	0.8312	0.7176	0.8576	0.9325
DHCNN	305	0.8038	0.7177	0.8162	0.9266	0.7889	0.6382	0.7948	0.9196	0.8407	0.8142	0.8827	0.9331
FAH	373	0.8223	0.7067	0.8359	0.9605	0.7976	0.6698	0.8151	0.9489	0.8562	0.8441	0.9029	0.9642
SCFR	421	0.8298	0.7194	0.8468	0.9794	0.8307	0.6422	0.8373	0.9626	0.8983	**0.8583**	0.9209	0.9879
CCLS2T (ours)	182	**0.9031**	**0.7353**	**0.9324**	**0.9876**	**0.9013**	**0.7315**	**0.9293**	**0.9715**	**0.9359**	0.8555	**0.9586**	**0.9955**

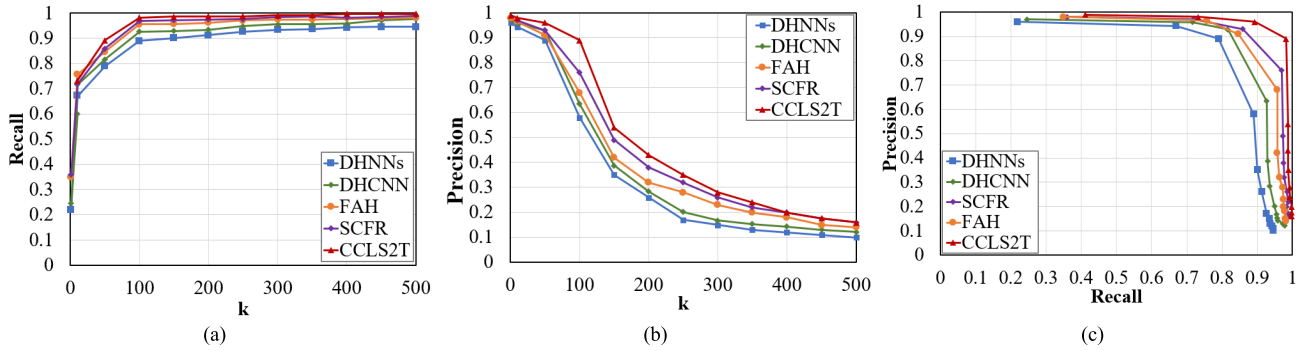


Fig. 9. Retrieval performance on the EuroSAT dataset. (a) Recall@k. (b) Precision@k. (c) Precision–recall.

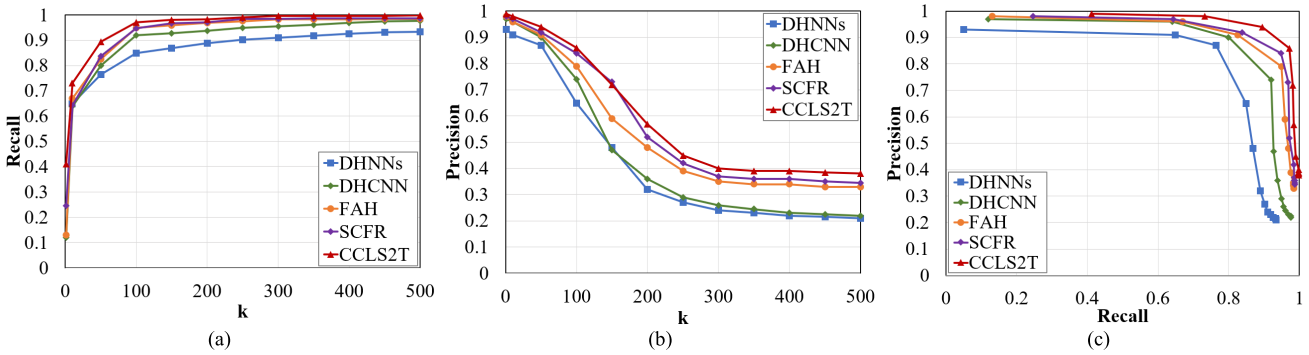


Fig. 10. Retrieval performance on the Million-AID dataset. (a) Recall@k. (b) Precision@k. (c) Precision–recall.

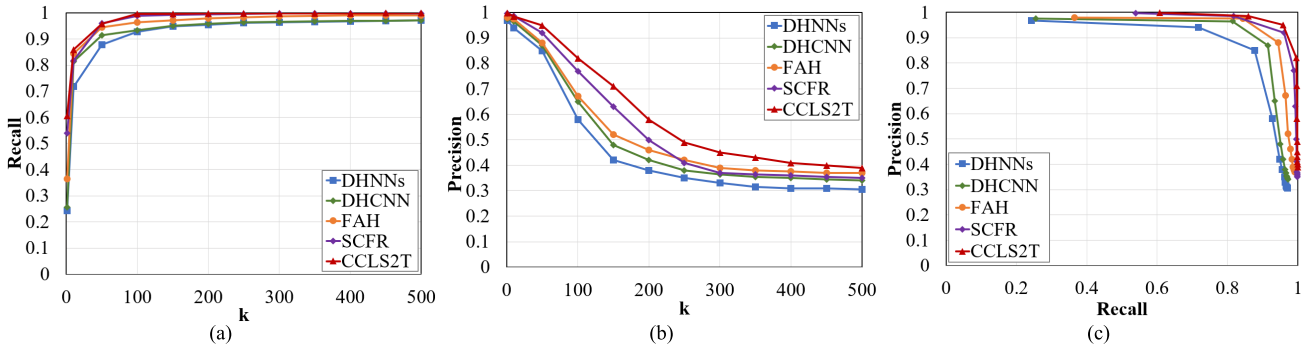


Fig. 11. Retrieval performance on the LivingAtlas dataset. (a) Recall@k. (b) Precision@k. (c) Precision–recall.

exhibit less-than-satisfactory performance in RSIR. In contrast, the FAH and SCFR methods surpass each other in terms of the recall@k and precision@k metrics. In contrast, our proposed CCLS2T outperforms the other methods on most occasions for the recall@k and precision@k metrics. Simultaneously, the precision–recall curve indicates the advantages of CCLS2T over other methods on different optical datasets. In accordance with the outcomes presented in Table II, the findings in Figs. 9–11 corroborate the superiority of our approach over the competing methods. The limitations of the DHNNs, DHCNN, FAH, and SCFR methods lie in their exclusive reliance on spatiotemporal information during the hash code generation process. Consequently, it is challenging for them to retrieve remote sensing images that are visually complex in an efficient manner, especially the images in the Million-AID dataset. Furthermore, the discrimination of remote sensing images from each other necessitates the integration of spatiotemporal information. In contrast, CCLS2T employs visual images and STC

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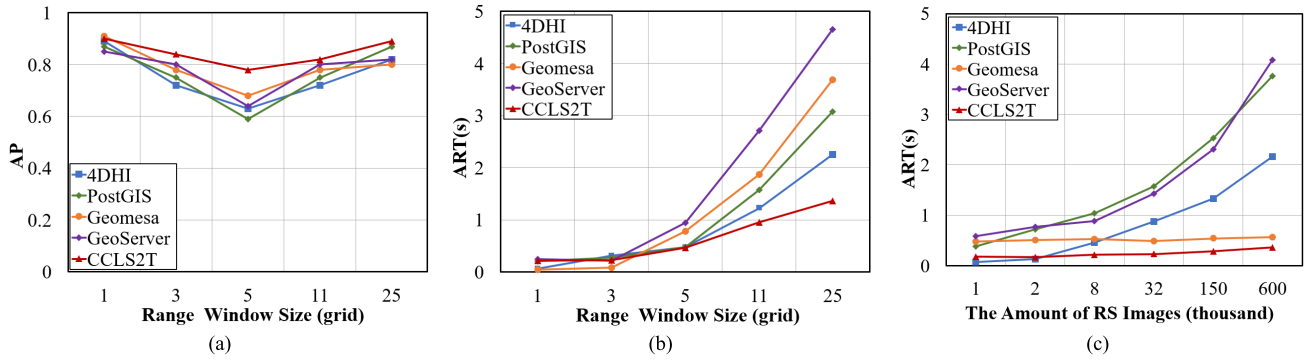


Fig. 12. Comparison results on RR. (a) AP on different range window sizes. (b) ART on different range window sizes. (c) ART on different amounts of remote sensing images.

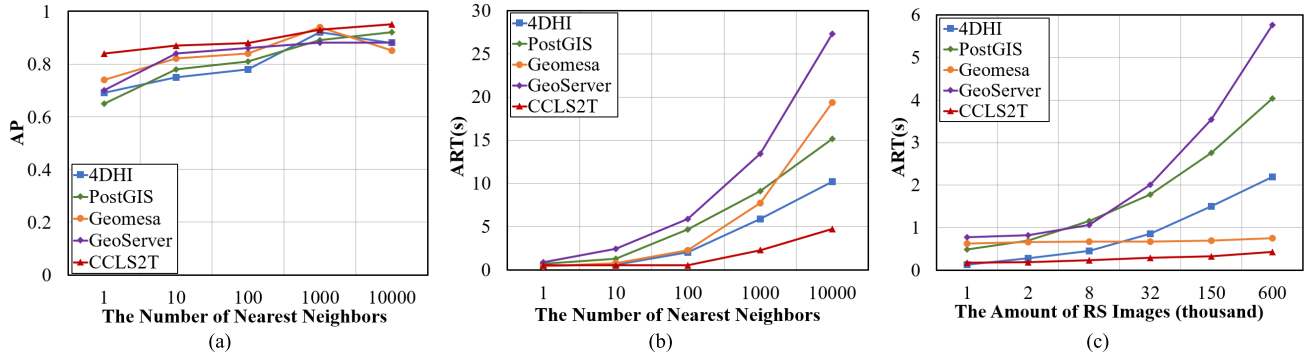


Fig. 13. Comparison results on KNN. (a) AP on different numbers of nearest neighbors in KNN. (b) ART on different number of nearest neighbors in KNN. (c) ART on different numbers of remote sensing images.

information to generate hash codes. This innovative approach enables CCLS2T to achieve high-RSIR accuracy. It demonstrates superior performance compared with that of the other algorithms, with an 11.64% higher AP rate. Therefore, our proposed method demonstrates a robust capacity for learning fine-grained semantics and spatiotemporal semantics, enabling effective breakthroughs beyond the existing approaches that prioritize visual feature extraction.

2) Comparison of the Efficiency of Other Server-Side Retrieval Engines: To validate the actual performance of CCLS2T, we perform efficiency experiments on two typical and real remote sensing image tasks, RR and KNN, from the perspectives of retrieval accuracy and retrieval efficiency. Compared service-side retrieval engines include the storage-side retrieval methods PostGIS and 4DHI and the service-side retrieval methods GeoNode and GeoMesa. The RR and KNN experimental results are given in Figs. 12 and 13, respectively.

Fig. 12 presents a comparison of results that vary with the size of the range window and the amount of data on the AP and ART metrics. According to the results in Fig. 12, with the expansion of the range window, the AP of all methods shows a trend of decreasing to the bottom first and then gradually increasing because a small range window corresponds to a small number of retrieval targets and a higher retrieval accuracy. As the range window increases, more redundant remote sensing images are returned, resulting in a decrease in retrieval accuracy. However, when the range window further increases, the redundant remote sensing images are concentrated on the boundary, and their influence gradually decreases, leading to a rebound in retrieval accuracy. Notably, in comparison to the other server-side retrieval engines, the proposed CCLS2T method demonstrates minimal changes in AP as the range window extends to a certain degree. For the RR efficiency, the ART of each method shows an obvious increasing trend with the expansion of the range window, where the ART growth of the GeoNode is much more conspicuous. Geomesa performs the best on RR with a small range window; however, it is also sensitive to the range window size, and a decrease in efficiency follows behind the GeoNode. Although our proposed CCLS2T may not be highly efficient when performing RR with a small range window, it stands out that the smoothness of ART changes as the range window expands. In addition, its efficiency decreases with the expansion of the range window but remains within reasonable limits when compared to other methods. In addition, the RR efficiencies of 4DHI, PostGIS, and GeoNode decrease significantly with increasing volume of remote sensing images. In contrast, that of GeoMesa is relatively stable. Our proposed CCLS2T has a stable RR efficiency and exhibits obvious advantages for different volumes of data.

In Fig. 13, the values of k were set to 1, 10, 100, 1000, and 10 000. As can be clearly seen from Fig. 13, all the methods show growth in AP as the value of k increases because the expansion of the similarity range in KNN makes it more likely that the correct remote sensing images are retrieved. PostGIS and 4DHI had relatively low APs in KNN, while CCLS2T had high APs in both KNN and above RR. From the perspective of KNN efficiency, there is a noteworthy increase in ART for each method as the value of k increases. When $k \leq 100$, an increase in k does not significantly affect query

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TABLE IV
RESULTS OF ABLATION STUDIES CONDUCTED ON THREE DATASETS

Ablation				EuroSAT				Million-AID				LivingAtlas			
STC	RSHST SMPL	IA	HNSM	AP	R@10	R@50	R@100	AP	R@10	R@50	R@100	AP	R@10	R@50	R@100
×	×	×	×	0.6379	0.5405	0.5436	0.5651	0.6172	0.6616	0.6567	0.6582	0.6107	0.6569	0.6307	0.6332
√	×	×	×	0.8354	0.8721	0.8762	0.8885	0.8265	0.8461	0.8428	0.8443	0.8689	0.8418	0.8751	0.8785
×	√	×	×	0.8556	0.9219	0.9229	0.9326	0.8275	0.9022	0.9147	0.9163	0.8249	0.9152	0.9212	0.9253
√	√	×	×	0.8633	0.9203	0.9201	0.9445	0.8703	0.9265	0.9275	0.9246	0.8366	0.9267	0.9307	0.9336
√	√	√	×	0.8857	**0.9356**	**0.9378**	0.9496	0.8933	0.931	0.9296	0.9289	0.8672	0.9294	0.9437	0.9467
√	√	√	√	**0.9024**	0.8905	0.9327	**0.9742**	**0.9129**	**0.9455**	**0.9442**	**0.9665**	**0.9008**	**0.9449**	**0.9565**	**0.9892**
×	√	√	√	0.7812	0.7531	0.7702	0.8021	0.8132	0.8381	0.8505	0.8691	0.8107	0.8474	0.8546	0.8544

efficiency, especially for the CCLS2T method, where the time consumption is mainly due to the local invocation process of the trained model. However, ART starts to increase when $k > 100$. Benefiting from the ability of CCLS2T to generate indexes with high accuracy, remote sensing image scanning is effectively limited to a small range, leading to a slower increase in ART. In terms of the relationship between retrieval efficiency and the number of remote sensing images, RR and KNN show similar trends. Consistent with the RR results, the KNN efficiency of 4DHI, PostGIS, and GeoNode decreases significantly with increasing volume of remote sensing images, while the efficiency of GeoMesa is relatively stable. In contrast, the proposed CCLS2T algorithm stably maintains the advantage of KNN efficiency on different volumes of remote sensing images.

The proposed CCLS2T algorithm generally achieves an average AP 9.91% greater than that of the other methods and maintains superior retrieval efficiency, demonstrating that CCLS2T outperforms server-side retrieval engines in terms of both accuracy and efficiency.

D. Ablation Studies

Ablation studies are performed on remote sensing IR to determine the impacts of the essential parts and operations outlined in CCLS2T. For our CCLS2T method, four key components—ITC learning and contrastive hash learning—are explored. The four components and operations are separately eliminated for ablation studies, including STC, RSHST-based sampling (RSHST SMPL) containing intraclass sampling and interclass sampling, IA, and HNSM. The detailed descriptions are as follows.

1) *STC:* It denotes whether the spatiotemporal context is introduced to construct an asymmetric dual-encoder architecture in CCLS2T.
2) *RSHST SMPL:* It indicates whether to adopt intraclass sampling or interclass sampling using the constructed RSHST.
3) *IA:* It indicates whether the IA method given in Section III-D1 is used for training.
4) *HNSM:* It indicates whether the HNSM given in Section III-D4 is used for training.

Table IV shows the results of ablation studies, where the AP, recall@10, recall@50, and recall@100 are used as metrics and are denoted as AP, R@10, R@50, and R@100, respectively. The bold numbers denote the best results for each column in each scenario. According to the results, it is clear that the STC and the RSHST SMPL have crucial effects on remote sensing image feature extraction, which effectively enhances the discriminative nature of the extracted features and significantly improves the retrieval accuracy (rows 1–3). In addition, the IA is also evidenced to promote further improvement in retrieval accuracy by effectively expanding the dataset size (rows 3 and 4). The HNSM operation allows the hash learning process to focus on learning the features of difficult-to-distinguish remote sensing images through dataset thinning and difficult sample supplementation, which greatly accelerates the convergence speed of the model.

Notably, the ablation experiment in terms of STC aimed to verify the outcomes of cross-modal contrastive learning in CCLS2T patients. As shown in Table IV, the retrieval accuracy is low when the cross-modal retrieval method is not used. When the STC is introduced, the retrieval experiments on all three datasets increase the retrieval accuracy. The finding that the retrieval accuracy decreases when RSHST SMPL, IA, and HNSM are adopted in CCLS2T simultaneously without STC (row 7) further corroborates the validity of cross-modal contrastive learning on images and STC of the proposed CCLS2T.

Subsequently, we assess the proposed techniques using various loss functions, and the outcomes are outlined in Table V. The cross-modal contrastive learning comparisons involve $\mathcal{L}_{\mathrm{CL}}$, $\mathcal{L}_{\mathrm{ITC}}$, and $\mathcal{L}_{\mathrm{ITM}}$, while contrastive hash learning comprises $\mathcal{L}^{I}_{\mathrm{CHL}}$, $\mathcal{L}^{E}_{\mathrm{CHL}}$, and $\mathcal{L}_{Q}$. In addition, Hard: $\mathcal{L}^{I}_{\mathrm{CHL}}$, $\mathrm{L}^{E}_{\mathrm{CHL}}$, and$\mathrm{L}_{Q}$ means applying HNSM in training dataset thinning and model optimization. The findings reveal that employing the complete loss function encompassing $\mathcal{L}_{\mathrm{CL}}$, $\mathcal{L}_{\mathrm{ITC}}$, and $\mathcal{L}_{\mathrm{ITM}}$ can enhance the discriminative capability of extracted features, which is a crucial factor in RSIR. Moreover, the results demonstrate that optimizing the network with $\mathcal{L}^{I}_{\mathrm{CHL}}$, $\mathcal{L}^{E}_{\mathrm{CHL}}$, and $\mathcal{L}_{Q}$ yields superior retrieval performance, leading to further improvement in retrieval accuracy. Although HNSM does not obviously enhance the retrieval accuracy, it contributes to accelerating the model training efficiency.

E. Parameter Analysis

It is widely known that the outcomes of a model are strongly influenced by hyperparameter values. We conducted additional experiments to verify the effect of these hyperparameters and their optimal values, including the RSHST construction parameters, loss function parameters, and hash code parameters.

For the RSHST construction parameters, we take η_v and η_{st} as two key hyperparameters to perform parameter analysis. η_v

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TABLE V
COMPARISON AMONG DIFFERENT LOSS FUNCTIONS FOR CCLS2T ON THREE DATASETS

Loss Function		EuroSAT				Million-AID				LivingAtlas			
Image-Text Contrastive Learning	Contrastive Hash Learning	AP	R@10	R@50	R@100	AP	R@10	R@50	R@100	AP	R@10	R@50	R@100
$\mathcal{L}_{CL}$	$\mathcal{L}^I_{CHL}$	0.8355	0.9112	0.9235	0.9254	0.8294	0.8332	0.9474	0.948	0.8279	0.8305	0.9205	0.9228
$\mathcal{L}_{CL}$	$\mathcal{L}^E_{CHL}$	0.8589	0.8776	0.8853	0.8901	0.8397	0.8459	0.8565	0.8592	0.8798	0.8821	0.8869	0.8876
$\mathcal{L}_{CL}$	$\mathcal{L}^I_{CHL}, \mathcal{L}^E_{CHL}$	0.8682	0.9187	0.9176	0.9341	0.8392	0.8447	0.9221	0.9247	0.8347	0.8372	0.9264	0.9279
$\mathcal{L}_{CL}$	$\mathcal{L}^I_{CHL}, \mathcal{L}^E_{CHL}, \mathcal{L}_Q$	0.9156	0.9432	**0.9274**	0.9573	**0.9103**	0.9142	**0.9487**	**0.9514**	**0.8956**	**0.8978**	**0.9512**	0.9524
$\mathcal{L}_{CL}$	Hard: $\mathcal{L}^I_{CHL}, \mathcal{L}^E_{CHL}, \mathcal{L}_Q$	**0.9167**	**0.9429**	0.9263	**0.9578**	0.9101	**0.9146**	0.9472	0.9505	0.8951	0.8967	0.9498	**0.9531**
$\mathcal{L}_{CL}, \mathcal{L}_{ITC}, \mathcal{L}_{ITM}$	$\mathcal{L}^I_{CHL}$	0.8642	0.8825	0.8907	0.8985	0.8462	0.8517	0.8619	0.8647	0.8882	0.8906	0.8937	0.8952
$\mathcal{L}_{CL}, \mathcal{L}_{ITC}, \mathcal{L}_{ITM}$	$\mathcal{L}^E_{CHL}$	0.8723	0.9132	0.9023	0.9315	0.8415	0.8457	0.8965	0.9002	0.8356	0.8389	0.9245	0.9278
$\mathcal{L}_{CL}, \mathcal{L}_{ITC}, \mathcal{L}_{ITM}$	$\mathcal{L}^I_{CHL}, \mathcal{L}^E_{CHL}$	0.9237	0.9465	0.9321	0.9624	0.9178	0.9234	0.9567	0.9591	0.9012	0.9038	0.9579	0.9587
$\mathcal{L}_{CL}, \mathcal{L}_{ITC}, \mathcal{L}_{ITM}$	$\mathcal{L}^I_{CHL}, \mathcal{L}^E_{CHL}, \mathcal{L}_Q$	**0.9295**	**0.9481**	0.9345	**0.9643**	**0.9188**	**0.9256**	0.9798	0.9823	0.9043	0.9065	0.9598	0.9612
$\mathcal{L}_{CL}, \mathcal{L}_{ITC}, \mathcal{L}_{ITM}$	Hard: $\mathcal{L}^I_{CHL}, \mathcal{L}^E_{CHL}, \mathcal{L}_Q$	0.9302	0.9478	**0.9352**	0.9639	0.9185	0.9249	**0.9805**	**0.9827**	**0.9049**	**0.9174**	**0.9601**	**0.9615**

TABLE VI
RSHST CONSTRUCTION PARAMETER EXPERIMENTS WITH DIFFERENT EPOCHS ON THE LIVINGATLAS DATASET

$\eta_v \backslash \eta_{st}$	0.01			0.03			0.05			0.07			0.1			0.3		
	DBI	h	AP	DBI	h	AP	DBI	h	AP	DBI	h	AP	DBI	h	AP	DBI	h	AP
0.01	3.7567	3	0.8557	2.7669	4	0.8321	1.7567	5	0.8756	0.7882	5	0.8507	0.7907	5	0.8102	5.8107	5	0.8022
0.03	1.5427	5	0.9632	1.8418	5	0.9634	1.2427	5	0.9773	0.8423	5	0.9651	0.8683	5	0.9057	3.8751	5	0.8941
0.05	0.8427	6	0.9880	0.3418	5	0.9915	0.0127	6	0.9945	0.2443	6	0.9534	0.7689	6	0.9261	1.0751	6	0.9074
0.07	1.9147	6	0.9801	1.5142	6	0.9829	0.9149	6	0.9987	1.3163	6	0.9833	2.8249	6	0.9305	2.9221	6	0.9143
0.1	3.2136	6	0.9545	2.8449	6	0.9406	1.9442	6	0.9658	3.9665	6	0.9613	4.9008	6	0.9157	5.9565	6	0.9016
0.3	5.9264	8	0.9107	3.9276	7	0.8833	2.9264	7	0.9232	7.9289	7	0.9215	9.8672	7	0.8720	8.9439	7	0.8645

and η_{st} work together to dynamically determine the threshold. These factors have significant repercussions for advancing the accuracy and efficiency of remote sensing data division in the RSHST and further affect the retrieval performance and training process of CCLS2T. In the experiments, η_v and η_{st} increase from 0.01 to 0.3 with progressively increasing steps. The corresponding results are given in Table VI, taking the DBI and the RSHST depth h as evaluation metrics. According to the results, with the gradual increase in η_v and η_{st}, DBI gradually approaches 0, indicating better classification in the RSHST with higher intraclass compactness and interclass separation. The constructed RSHST further affects the training process of CCLS2T through intraclass and interclass sampling and HNSM, which have certain impacts on its overall retrieval effect. When η_v and η_{st} are in the range of [0.05, 0.07], DBIs are closest to 0, implying that the RSHST construction results are optimal at this time and that the corresponding CCLS2T retrieval accuracy is also the highest. Thereafter, with further increases in the two parameters η_v and η_{st}, the DBI of RSHST starts to increase, which implies that the construction effect of RSHST starts to decrease; consequently, the retrieval accuracy of CCLS2T decreases again.

For the loss function parameters, we take λ_{ITC} and λ_{ITM} as the key hyperparameter for performing parameter analysis; these parameters are used to regulate the contribution of the classification loss $\mathcal{L}_{\text{CL}}$, the ITC loss $\mathcal{L}_{\text{ITC}}$, and the ITM loss function $\mathcal{L}_{\text{ITM}}$ in $\mathcal{L}_{\text{DFE}}$. The values of λ_{ITC} and λ_{ITM} are in the range of $[0.1, 0.9]$, with an increasing step of 0.1. The experimental results of the loss function weight sensitivity in $\mathcal{L}_{\text{DFE}}$ on three optical datasets are given in Table VII. According to the results, as λ_{ITC} and λ_{ITM} increase, the retrieval accuracy gradually increases. When they are in the range of $[0.4, 0.5]$, CCLS2T obtains more distinguished features and relatively better retrieval performance. When their values continue to increase, the performance of CCLS2T decreases again.

TABLE VII
R@100 OF LOSS FUNCTION PARAMETER EXPERIMENT IN $\mathcal{L}_{\text{DFE}}$ ON THE LIVINGATLAS DATASET

$\lambda_{ITC} \backslash \lambda_{ITM}$	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
0.1	0.7648	0.8159	0.8567	0.8547	0.8581	0.8314	0.8255	0.8095	0.8105
0.2	0.766	0.8191	0.8582	0.8626	0.8632	0.8406	0.8618	0.8379	0.8411
0.3	0.8357	0.8425	0.8881	0.8944	0.8452	0.8488	0.8515	0.8872	0.8829
0.4	0.9061	0.9394	0.9748	0.9625	0.9529	0.9402	0.9314	0.9260	0.9140
0.5	0.9144	0.9464	0.9906	0.9755	0.969	0.9454	0.945	0.9068	0.9102
0.6	0.8746	0.9284	0.9361	0.9492	0.9276	0.9272	0.9262	0.913	0.8972
0.7	0.8693	0.9222	0.9312	0.9439	0.9241	0.922	0.9192	0.899	0.8811
0.8	0.8278	0.919	0.9255	0.933	0.9164	0.9151	0.9047	0.8848	0.8665
0.9	0.8267	0.8493	0.8843	0.8899	0.8454	0.8469	0.8463	0.8391	0.8375

For hash code parameters, we perform parameter analysis on the hash code length H. Its impact on model performance is mainly reflected in the fact that longer hash codes are beneficial for differentiating remote sensing images; in contrast, an increase in length also brings about an increase in model size and training cost. As shown in Table VIII, the corresponding retrieval accuracies are verified on different optical datasets with generated hashing indexes of 32, 64, 96, 128, 256, and 512. Analysis of Table VIII shows that with increasing hash code length, the retrieval accuracy gradually increases. However, once the hash code length exceeds 256 bits, the retrieval accuracy tends to be stable and even shows a downward trend on some datasets. Simultaneously, as the hash code length

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TABLE VIII
R@100 OF HASH CODE PARAMETER EXPERIMENT WITH DIFFERENT EPOCHS ON THREE DATASETS

H\Epochs	EuroSAT				Million-AID				LivingAtlas			
	50	100	200	300	50	100	200	300	50	100	200	300
32	0.5506	0.6017	0.6273	0.6548	0.5714	0.6367	0.6526	0.6834	0.5902	0.6565	0.6708	0.7042
64	0.6558	0.6951	0.7115	0.7481	0.6872	0.7264	0.7448	0.7709	0.7013	0.7414	0.7609	0.7991
96	0.7879	0.9519	0.9731	0.9875	0.8147	0.9516	0.9784	0.9965	0.8205	0.9585	0.9795	0.9962
128	0.8823	0.9917	0.9893	0.9982	0.8973	0.9942	0.9969	0.9998	0.9026	0.9954	0.9971	0.9978
256	0.8818	0.9569	0.9847	0.9984	0.8964	0.9667	0.9853	0.9986	0.9019	0.9479	0.9864	0.9976
512	0.8725	0.9453	0.9732	0.9979	0.8877	0.9401	0.9783	0.9974	0.8926	0.9218	0.9695	0.9877

increases, additional epochs are needed to train the model to the optimal performance, indicating a decrease in the training efficiency. However, once the code length reaches a certain threshold, the retrieval time starts to increase again due to the reduction in the scanning scope resulting from the increased index length. Nevertheless, excessively long index lengths can impose additional pressure on reading and writing operations.

V. CONCLUSION

This article proposes CCLS2T, a new framework for MSRSIR with cross-modal contrastive learning. In addition to feature representation, CCLS2T is more concerned with the mining of semantic correlations as hash-based indexing from multisource optical remote sensing datasets. Inspired by recent popular image–text pretraining models, we develop two methods to improve the RSIR performance. First, extensive spatiotemporal remote sensing contexts are reconstructed (R2STFT) and embedded in contrastive learning as geographic knowledge to improve the discrimination of multisource images with similar global visual features. Moreover, an RSHST is designed to increase the multiple-feature correlation of remote sensing images and generate new sample pairs for contrastive hash learning. We theoretically and experimentally validate the effectiveness of the proposed ITC learning and hash-based retrieval methods. Compared with existing methods, CCLS2T performs better on various retrieval subtasks.

Although this article offers encouraging results on vision-language contrastive learning for retrieval performance, supplementary analysis of the method is necessary in the future. We would like to provide more prompt templates for integrating heterogeneous remote sensing-based metadata corpora rather than spatiotemporal texts. In addition, we can explore more elements for correlation formulation to enrich the index semantic instructor. Moreover, we should not only focus on constructing or generating new index blocks but also delve into how large-scale historical indexes can be effectively updated online in real time.

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